

Capacity of a constrained urban airspace: Influencing factors, analytical modelling and simulations

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ABSTRACT

The traffic density of small aerial vehicles operating within urban environments is expected to increase significantly in the near future. This urban environment is highly constrained due to being limited to the low-altitude airspace directly above the existing road network. Multiple studies have addressed factors influencing the capacity of urban airspace. These have used simulations of aircraft, yet the empirical nature of these simulations limits their use beyond the specific conditions that have been tested. Analytical models would not have this limitation, but they are only developed for general airspace, while the emergent patterns in constrained urban airspace are different than in general, unconstrained conditions. For instance, queuing and local congestion are patterns that are unique to the heavily-constrained environment. Therefore, in this paper, we derive an analytical model for air traffic in a confined airspace to find the influencing factors for its capacity. By means of a simulation of aerial vehicles, we verify the analytical model and show a relationship between the mean flow rate and mean density in a two-dimensional orthogonal grid network airspace. Results show that the entire airspace can become unstable when the maximum capacity of just one intersection is reached. Furthermore, the maximum airspace density is found to be unaffected by cruise speed. The results demonstrate how the derived analytical model provides an effective tool to predict the effect of several design parameters on the capacity of constrained urban airspace. Moreover, this model can form the basis for further extensions, including the altitude dimension and non-orthogonal or non-four-way intersections.

1. Introduction

In recent years, the development of unmanned aerial vehicles (UAVs) and personal aerial vehicles (PAVs) have been proposed as solutions for the increasing number of road congestion problems within cities. Doole et al. (2020) estimated that, even for a conservative growth rate, the traffic density of parcel and food delivery UAVs in the urban airspace of Paris would reach more than 60 thousand simultaneously operating vehicles by 2035. These numbers are substantially higher than current en-route airspace densities for conventional aviation (Hoekstra et al., 2000). Furthermore, the environment of the low-altitude urban airspace is highly constrained. Flying above existing infrastructures, such as roads and waterways, is desired to reduce exposure to the urban population (Salleh et al., 2018; Doole et al., 2021) or even unavoidable due to the presence of high buildings in the low-altitude

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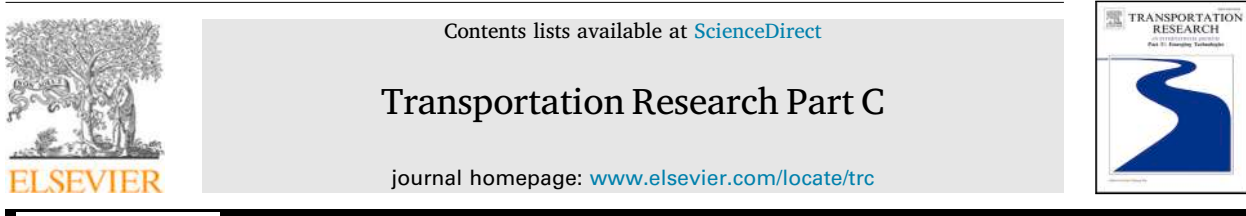
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受限城市空域容量：影响因素、分析建模与仿真

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城市交通管理（UTM）
蓝天自动终端情报服务模拟器
交通仿真

摘要

预计在不久的将来，城市环境中运行的小型飞行器的交通密度将显著增加。由于仅限于现有道路网络正上方的低空空域，这种城市环境受到高度限制。多项研究探讨了影响城市空域容量的因素。这些研究使用了飞机模拟，但这些模拟的经验性质限制了它们在已测试的特定条件之外的应用。分析模型不会有这个限制，但它们仅针对一般空域开发，而受限城市空域中出现的模式与一般的、不受限的条件不同。例如，排队和局部拥堵是高度受限环境所特有的模式。因此，在本文中，我们推导了受限空域中空中交通的分析模型，以找出影响其容量的因素。通过飞行器模拟，我们验证了分析模型，并展示了二维正交网格网络空域中平均流量与平均密度之间的关系。结果表明，当仅一个交叉路口达到最大容量时，整个空域可能会变得不稳定。此外，发现最大空域密度不受巡航速度的影响。结果表明，所推导的分析模型如何提供一个有效的工具来预测几个设计参数对受限城市空域容量的影响。此外，该模型可以作为进一步扩展的基础，包括高度维度以及非正交或非四路交叉路口。

1. 引言

近年来，无人驾驶飞行器（UAV）和个人飞行器（PAV）的发展被提议作为解决城市道路拥堵问题日益增多的方案。Doole等人（2020年）估计，即使以保守的增长率计算，到2035年，巴黎城市空域中包裹和食品配送无人机的交通密度将达到同时运行超过6万辆的车辆。这些数字大大高于当前常规航空的航路空域密度（Hoekstra等人，2000年）。此外，低空城市空域的环境受到高度限制。在现有基础设施（如道路和水道）上方飞行，希望减少对城市人口的影响（Salleh等人，2018年；Doole等人，2021年），或者由于低空存在高层建筑甚至是不可避免的。

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Nomenclature	
a	acceleration or deceleration capability [m/s ²]
c	cycle time of a traffic signal [s]
D	flight distance, or leg length [m]
d	delay [s]
F	set of all legs f at a single intersection
k	comparison scaling parameter [m]
L	queue length [m]
N_i	number of instantaneous aircraft [-]
p	proportion of aircraft passages [-]
q	flow rate [s ⁻¹]
r	red time of a traffic signal [s]
S	conflict-free flight distance between two aircraft [m]
S_h	horizontal separation requirement [m]
s	intersection flow capacity [s ⁻¹]
t	time [s]
V	aircraft speed [m/s]
x	degree of saturation [-]
Z	set of all network legs z
Superscripts:	
-	mean value of
Subscripts:	
c	of the crossing aircraft
d	downstream intersection leg
E	eastbound through flow
EN	east- to northbound turning flow
f	intersection leg
g	general (delay)
i	inflow (rate)
l	look-ahead
N	northbound through flow
NE	north- to eastbound turning flow
NR	without (no) conflict resolution
s	stochastic (delay)
u	upstream intersection leg
z	network leg
λ	proportion of time an intersection is free to cross
ψ	ground track angle [rad]
+	crossing at right angles
∠	crossing at an angle
	originating from the same upstream leg
%	relative (comparison scaling parameter) [%]

urban airspace. The high level of aircraft density and the increased level of constraints are not found in regular, en-route airspace, raising the question of whether and how the constrained urban airspace can sustain such densities.

In previous studies, it has been shown that imposing some structure on an airspace can have beneficial effects on capacity and safety. Sunil et al. (2015) and Sunil et al. (2017) conclude that non-constrained airspace capacity benefits from a limited degree of structuring, namely, adding altitude layers with a reduced heading range. This vertical segmentation was confirmed to be beneficial as well for the constrained urban airspace (Doole et al., 2021). The latter study also demonstrates the advantages of enforcing unidirectional flows along streets, as opposed to allowing both directions (at different altitudes). Similarly, Salleh et al. (2017) argued that a structured airspace design is a prerequisite to enable large-scale urban airspace operations, with follow-up research showing that the airspace should not be overly constrained (Salleh et al., 2018). In all studies mentioned, simulations are used to

术语表	
a	加速或减速能力 [m/s ²]
c	交通信号周期时间 [秒]
D	飞行距离，或航段长度 [米]
d	延误 [秒]
F	单个交叉路口处所有航段 f 的集合
k	比较缩放参数 [m]
L	队列长度 [m]
N_i	瞬时飞行器数量 [-]
p	飞行器通过比例 [-]
q	流量 [s ⁻¹]
r	交通信号的红灯时间 [秒]
S	两架飞机之间无冲突飞行距离 [米]
S_h	水平间隔要求 [米]
S	交叉路口流量 [s ⁻¹]
t	时间 [秒]
V	飞机速度 [m/s]
x	饱和度 [-]
Z	所有网络路段的集合 z
上标:	
-	……的平均值
下标:	
c	交叉飞机的
d	下游交叉路口路段
E	东向直通流量
EN	东向北向转弯流量
f	交叉路口路段
g	一般情况（延误）
i	流入（速率）
l	前瞻
N	北向直流通行
NE	北向至东向转向流
NR	无（不进行）冲突解决
S	随机的（延迟）
u	上游交叉路段
z	网络航段
λ	交叉路口可自由通行的时间比例
ψ	航迹角 [弧度]
+	直角交叉
∠	以一定角度交叉
	源自同一条上游路段
%	相对（比较缩放参数） [%]

城市空域。在常规的航路空域中，并未出现如此高的飞机密度和增加的限制水平，这就引发了一个问题：受限的城市空域能否以及如何维持这样的密度。在先前的研究中，已经表明对空域施加某种结构可以对容量和安全产生有益影响。Sunil等人（2015年）和Sunil等人（2017年）得出结论，无约束空域容量受益于有限程度的结构化，即增加航向范围减小的高度层。这种垂直分割对于受限的城市空域也被证实是有益的（Doole等人，2021年）。后一项研究还证明了沿街道强制单向流动的优势，而不是允许两个方向（在不同高度）。同样，Salleh等人（2017年）认为结构化空域设计是实现大规模城市空域运行的先决条件，后续研究表明空域不应受到过度约束（Salleh等人，2018年）。在所有提及的研究中，都使用模拟来

compare the different concepts for airspace structure. With simulations, we refer to a computer simulation of individual aircraft interacting with each other, in which their movements are determined by computer code (no human in the loop). In different fields, this is known under various names: fast-time simulations, agent-based simulations or microscopic simulations. In this paper, we will simply use the word “simulations” for this.

Up to now, the simulations are empirical in nature (i.e., they are applied to specific conditions), which limits their use beyond the specific conditions tested (Sunil et al., 2018c). Notably, restricted airspaces are not well studied. At the moment, it is unknown which parameters influence traffic operations or road capacity. This is addressed in the current paper. We follow the approach suggested by Sunil et al. (2018b,a) to create a mathematical model using an analytical approach that can quantify the factors influencing the capacity of constrained urban airspace. The suggested approach is based on conflict count models and domino effect models as predictors of airspace capacity.

This study will hence investigate the factors that influence the capacity of constrained urban airspace. An analytical method is proposed that estimates the relationship between the mean inflow rate and the mean density, for which we use a two-dimensional, orthogonal grid network airspace. As a result, the effect of several airspace design parameters on constrained urban airspace capacity can be quantified. These insights can be used in urban airspace design applications. They can also provide a basis for the derivation of more expanded models, enabling an increased comprehension and modelling of the constrained urban airspace.

This paper continues by presenting the techniques available to model airspace capacity, followed by the definitions and technical terminology we will be using in Section 3. Subsequently, Section 4 elaborates on the interactions between aircraft in a constrained urban environment, laying the groundwork for the derivation of the analytical model in Section 5. Next, by means of a simulation experiment, the level of accuracy of the analytical model is assessed. The experimental setup thereof is discussed in Section 6, and the empirical results of these simulation experiments are compared to the predictions of the analytical model in Section 7. In Section 8, we discuss the level of accuracy of the derived model, its limitations, and key findings and limitations of this study. Additionally, opportunities for future research are presented. Finally, the main conclusions are summarised in Section 9.

2. Models for airspace capacity

For free, unconstrained airspace, several analytical models of airspace capacity exist. These models relate capacity to airspace stability and conflict probability, using binomial combinatorics-based equations. They are typically referred to as ‘gas model’ equations and have been proposed in aviation research since the late 1960s (Reich, 1966a,b; Flanagan and Willis, 1969; Endoh, 1982; Brooker, 2002; Jardin, 2005; Netjasov, 2012; Sunil et al., 2018b,a). These also are proposed as dynamic discrete-time simulation models (e.g., Gharibi et al., 2019). They have various applications, including traffic control (e.g., Zhou et al., 2021; Haddad et al., 2021). These models rely on several assumptions, such as a uniform distribution of traffic and flight direction, which make them not suitable for constrained urban airspace.

In a constrained environment, the limited horizontal manoeuvre space prevents lateral resolution manoeuvres. Moreover, when traffic is distributed across altitude layers of different speed or heading directions, as advocated by Sunil et al. (2015, 2017), Jang et al. (2017), Doole et al. (2021), vertical resolution possibilities are limited. As a result, of the three available en-route conflict resolution options (Kuchar and Yang, 2000), this paper will focus on speed-based algorithms. Similar to the analytical models of Sunil et al. (2018b,a), strategic conflict resolution, also called pre-departure conflict resolution, will not be considered. Therefore, importantly, in this paper we consider tactical, en-route conflict resolution to be performed using speed-based algorithms.

An emergent behaviour resulting from speed-based resolution manoeuvres in a constrained environment is that aircraft queue upstream of busy intersections. In other words, within an airway, an aircraft can be in continuous conflict with the aircraft directly in front of it. The essentially one-dimensional nature of traffic following a street will cause knock-on conflicts to be much more commonplace, which reduces the value of conflict count models and domino effect models as predictors of airspace capacity. Furthermore, as concluded by Sunil et al. (2018c), the accuracy of their earlier proposed analytical model (Sunil et al., 2018b) degrades when creating artificial ‘hotspots’ with a higher traffic density. Constrained urban airspace is a clear example of an environment with many hotspots, as traffic can become concentrated on busy streets and intersections. Altogether, the existing analytical models are not valid for constrained urban airspace, signifying the need for a tailored analytical model.

While constrained urban airspace has different characteristics than non-constrained airspace, it has many similarities with road transportation environments. This makes road transportation research an obvious inspiration for capacity modelling of constrained urban airspace. Similar to our observation above about knock-on conflicts in streets, road transportation research also seldom considers conflicts. Instead, network efficiency and instability are extensively investigated. Recent research in urban road transportation has focused on the relationship between the mean inflow rate and the mean density in a network (Geroliminis and Daganzo, 2008; Daganzo et al., 2011; Gayah and Daganzo, 2011; Knoop et al., 2012). This relationship is expressed in the so-called Macroscopic Fundamental Diagram (MFD), which has been experimentally confirmed in a study on actual road traffic in Yokohoma, Japan (Geroliminis and Daganzo, 2008) and various other cities since. See Loder et al. (2019) for a comparison of 41 cities. Using the MFD, the behaviour of a road network can be observed, such that the traffic status (free-flow or congestion), critical density, and efficiency are identified.

As aircraft behaviour shows many similarities to road traffic behaviour in constrained urban environments, the adaptation of the Macroscopic Fundamental Diagram to assess constrained urban airspace has potential and will hence be explored here. This has had some first exploration based on particular conflict avoidance using simulations (e.g., Jang et al., 2017; Cummings and Mahmassani, 2021; Safadi et al., 2023). In the current paper, we will combine simulations and analytical approaches for constraint environments. Furthermore, we will focus on the free-flow part and the critical density point of the MFD. The congestion part is assumed less relevant for air traffic operations, as the regulated inflow means congestion can be avoided.

比较不同的空域结构概念。通过模拟，我们指的是单个飞机相互作用的计算机模拟，其中它们的运动由计算机代码确定（无人工干预）。在不同领域，这有各种名称：快速时间模拟、基于代理的模拟或微观模拟。在本文中，我们将简单地使用“模拟”这个词来指代。

到目前为止，模拟本质上是经验性的（即它们应用于特定条件），这限制了它们在测试的特定条件之外的使用（Sunil等人，2018c）。值得注意的是，受限空域没有得到很好的研究。目前，尚不清楚哪些参数会影响交通运行或道路容量。本文将解决这个问题。我们遵循Sunil等人（2018b, a）建议的方法，使用分析方法创建一个数学模型，该模型可以量化影响受限城市空域容量的因素。建议的方法基于冲突计数模型和多米诺效应模型作为空域容量的预测器。

因此，本研究将调查影响受限城市空域容量的因素。提出了一种分析方法，用于估计平均流入率与平均密度之间的关系，为此我们使用二维正交网格网络空域。结果，可以量化几个空域设计参数对受限城市空域容量的影响。这些见解可用于城市空域设计应用。它们还可以为推导更扩展的模型提供基础，从而能够更好地理解和建模受限城市空域。

本文接下来将介绍可用于建模空域容量的技术，随后是我们将在第3节中使用的定义和技术术语。随后，第4节详细阐述受限城市环境中飞机之间的相互作用，为第5节中分析模型的推导奠定基础。接下来，通过模拟实验评估分析模型的准确性水平。第6节讨论其实验设置，第7节将这些模拟实验的实证结果与分析模型的预测进行比较。在第8节中，我们讨论推导模型的准确性水平、其局限性以及本研究的主要发现和局限性。此外，还提出了未来研究的机会。最后，第9节总结主要结论。

2. 空域容量模型

对于自由的、无约束的空域，存在几种空域容量的分析模型。这些模型使用基于二项式组合的方程将容量与空域稳定性和冲突概率联系起来。它们通常被称为“气体模型”方程，自20世纪60年代末以来已在航空研究中提出（Reich, 1966a, b; Flanagan和Willis, 1969; Endoh, 1982; Brooker, 2002; Jardin, 2005; Netjasov, 2nd12; Sunil等人，2018b, a）。这些也被提出作为动态离散时间模拟模型（例如，Gharibi等人，2019）。它们有各种应用，包括交通控制（例如，Zhou等人，2021; Haddad等人，2021）。这些模型依赖于几个假设，例如交通和飞行方向的均匀分布，这使得它们不适用于受限的城市空域。

在受限环境中，有限水平机动空间阻碍横向冲突解决。此外，正如Sunil等人（2015、2017）、Jang等人（2017）、Doole等人（2021）所主张，当交通分布在不同速度或航向高度层时，垂直冲突解决可能性也受限。因此，在三种可用航路冲突解决选项中（Kuchar和Yang, 2000），本文将重点关注基于速度的算法。与Sunil等人（2018b, a）分析模型类似，战略冲突解决（出发前冲突解决）将不被考虑。因此，本文认为战术性航路冲突解决将使用基于速度的算法执行。

在受限环境中，基于速度的冲突解决机动产生的一个突发行是，飞机在繁忙交叉路口上游排队。换句话说，在一条航路内，一架飞机可能会与直接在其前方的飞机持续发生冲突。沿着街道行驶的交通本质上的一维性质将导致连锁冲突更加普遍，这降低了冲突计数模型和多米诺效应模型作为空域容量预测指标的价值。此外，正如Sunil等人（2018c）所总结的，当创建具有更高交通密度的人工“热点”时，他们早期提出的分析模型（Sunil等人，2018b）的准确性会下降。受限城市空域就是一个有许多热点的环境的明显例子，因为交通可能会集中在繁忙的街道和交叉路口。总之，现有的分析模型对受限城市空域无效，这表明需要一个量身定制的分析模型。

虽然受限城市空域与非受限空域具有不同的特征，但它与道路交通环境有许多相似之处。这使得道路交通研究成为受限城市空域容量建模的一个明显灵感来源。与我们上面关于街道连锁冲突的观察类似，道路交通研究也很少考虑冲突。相反，网络效率和不稳定性得到了广泛研究。城市道路交通的最新研究集中在网络中的平均流入率和平均密度之间的关系上（Gerolimimis和Daganzo, 2008年；Daganzo等人，2011年；Gayah和Daganzo, 2011年；Knoop等人，2012年）。这种关系在所谓的宏观基本图（MFD）中得到了体现，该图已在日本横滨的实际道路交通研究中（Gerolimimis和Daganzo, 2008年）以及此后的其他各个城市中得到了实验验证。有关41个城市的比较，请参见Loder等人（2019年）。使用宏观基本图，可以观察道路网络的行为，从而确定交通状态（自由流或拥堵）、临界密度和效率。

由于在受限城市环境中飞机行为与道路交通行为有许多相似之处，因此将宏观基本图应用于评估受限城市空域具有潜力，因此本文将对此进行探索。此前已经基于特定的冲突避免使用模拟进行了一些初步探索（例如，Jang等人，2017年；Cummings和Mahmassani, 2021年；Safadi等人，2023年）。在本文中，我们将结合模拟和分析方法来研究受限环境。此外，我们将关注宏观基本图的自由流部分和临界密度点。由于管制的流入意味着可以避免拥堵，因此拥堵部分被认为与空中交通运营的相关性较小。

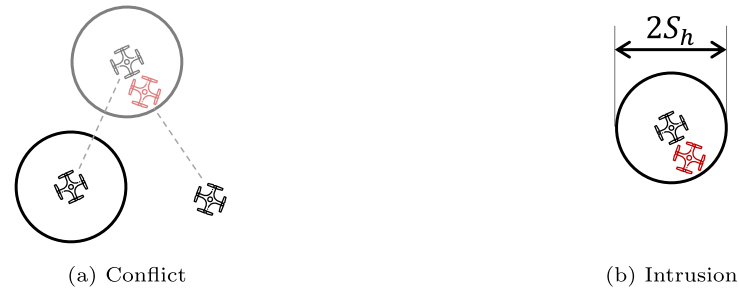


Fig. 1. The difference between a conflict, i.e., a predicted, future intrusion, and an intrusion in the horizontal plane, with S_h the horizontal separation requirement.

3. Conflicts, intrusions and their resolution

This section explains conflicts, intrusions, and the tactical conflict resolution methods used in this paper.

An *intrusion*, also called loss of separation, occurs when two aircraft simultaneously violate the minimum horizontal and vertical separation requirements. In this paper, an aircraft is assumed to be any small flying vehicle with hovering capabilities, e.g., UAVs, PAVs, or helicopters. When an intrusion is predicted to occur within a predetermined look-ahead time, this is called a *conflict*. The difference between conflicts and intrusions is shown in Fig. 1. By definition, conflicts and intrusions are pairwise, i.e., between two aircraft only.

To prevent intrusions, most research in urban airspace either makes use of pre-departure separation (Salleh et al., 2018; Peinecke and Kuenz, 2017) or of en-route decentralised air traffic management (ATM) performed by each aircraft autonomously (Sunil et al., 2015, 2017; Doole et al., 2021). The model derivations in this study assume decentralised tactical separation management, based on state-based *conflict detection* and speed-based *conflict resolution*. With state-based conflict detection, aircraft positions are linearly extrapolated along the velocity vector to detect conflicts within a given look-ahead time. Note that, when using state-based detection, changes in state, such as turns and accelerations are not predicted. It is assumed, however, that aircraft have perfect knowledge of their own and others' current state. In reality, the accuracy of this state information can be considered in the design choice for the horizontal and vertical separation requirements, to prevent inaccuracies from resulting in collisions.

Upon detection of a conflict, an aircraft will perform tactical conflict resolution to avoid the predicted intrusion. As mentioned in the introduction, speed-based conflict resolution algorithms are used. The implementation of speed-based conflict resolution means that airways can be one-way only (per altitude level), as head-on conflicts are not resolvable.

Moreover, analogous to road transportation networks, speed limits are possibly applied in specific airways or locations. It is assumed that all aircraft will fly at equal cruise speed V_{NR} when no conflict resolution (NR) would be used, likely close to such a speed limit. Already, the FAA proposed a speed limit for small UAVs of 45 m/s (Federal Aviation Administration, 2015). In addition, the International Rules of the Air (International Civil Aviation Organization, 2005) define that an overtaking aircraft has to give right-of-way to the aircraft being overtaken. Due to both reasons, the resolution of a conflict by accelerating is assumed infeasible in a two-dimensional constrained environment. Hence, the speed with conflict resolution (WR) V_{WR} will always be lower than the cruise speed. By assuming hovering capabilities for all aircraft, the lower bound equals 0, and the range of V_{WR} is:

$$V_{WR} \in [0, V_{NR}] \quad (1)$$

For conflicts within an airway, the speed-based algorithm used follows the International Rules of the Air (International Civil Aviation Organization, 2005), dictating that an aircraft overtaking another aircraft needs to give way. For crossing conflicts, the aircraft which has to incur the slightest delay will perform the resolution. This aircraft is the aircraft that crosses on the rear-side of the other aircraft, i.e., the aircraft which, at the closest point of approach, has a relative bearing with the other aircraft in the range of $[-90^\circ, 90^\circ]$.

4. Aircraft interactions in a constrained environment

In a constrained airspace, the interactions between aircraft differ from the interactions encountered in a non-constrained environment. By analysing these interactions, several assumptions can be made which facilitate the derivation of an analytical model for the relationship between the mean inflow rate and the aircraft density. These assumptions relate to the maximum throughput capacity of a single intersection, which is an input for the analytical model. Additionally, the theoretical upper bounds for the aircraft density are set by the assumptions made in this section.

This section presents several aircraft interactions that can be identified in a two-dimensional (i.e., all aircraft fly at the same altitude level) constrained urban airspace environment. For demonstration purposes, an orthogonal intersection of two one-way airways will be used. At this single orthogonal intersection, two types of traffic are distinguished. *Through traffic* represents the set of aircraft that continues on a straight path, whereas *turning traffic* represents the set of aircraft that turns onto the other airway. Firstly, the effect of the interaction between through traffic of two flows crossing a single intersection is evaluated. Next, the additional

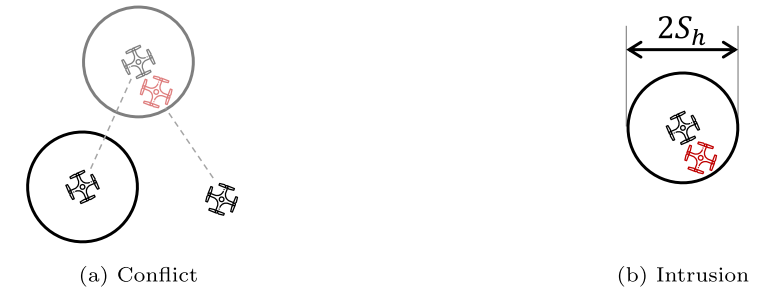


图1. 冲突（即预测的未来侵入）与水平面内侵入之间的区别，其中 S_h 为水平间隔要求。

3. 冲突、侵入及其解决

本节解释冲突、侵入以及本文中使用的战术性冲突解决方法。

侵入，也称为间隔丧失，当两架飞机同时违反最小水平和垂直间隔要求时发生。在本文中，飞机被假定为任何具有悬停能力的小型飞行器，例如无人机、个人飞行器或直升机。当预计在预定的前瞻时间内会发生侵入时，这称为冲突。冲突和侵入之间的区别如图1所示。根据定义，冲突和侵入是成对的，即仅在两架飞机之间。

为防止入侵，城市空域的大多数研究要么采用离场前间隔（Salleh等人，2018年；Peinecke和Kuenz，2017年），要么采用每架飞机自主进行的航路分散式空中交通管理（ATM）（Sunil等人，2015年、2017年；Doole等人，2021年）。本研究中的模型推导假设基于基于状态的冲突检测和基于速度的冲突解决的分散式战术间隔管理。通过基于状态的冲突检测，沿速度矢量线性外推飞机位置以检测给定前瞻时间内的冲突。请注意，使用基于状态的检测时，不会预测状态变化，如转弯和加速。然而，假设飞机完全了解自身和其他飞机的当前状态。在实际中，可在水平和垂直间隔要求的设计选择中考虑此状态信息的准确性，以防止不准确导致碰撞。

检测到冲突后，飞机会执行战术冲突解决以避免预测的入侵。如引言中所述，使用基于速度的冲突解决算法。基于速度的冲突解决的实施意味着航路只能是单向的（每个高度层），因为迎头冲突无法解决。

类似于道路交通网络，特定航路或位置可能应用速度限制。假设不使用冲突解决(NR)时，所有飞机以相等巡航速度 V_{NR} 飞行，可能接近此限制。美国联邦航空管理局(2015年)已为小型无人机提议45米/秒速度限制。《国际空中规则》(ICAO, 2005年)规定超飞机必须给被超飞机让路。因此，在二维受限环境中通过加速解决冲突不可行，冲突解决速度 V_{WR} 将低于巡航速度。假设所有飞机具有悬停能力，下限为0， V_{WR} 范围为：

$$V_{WR} \in [0, V_{NR}] \quad (1)$$

对于航路内的冲突，所使用的基于速度的算法遵循《国际空中规则》（International Civil Aviation Organization, 2005年），规定超车飞机需要让路。对于交叉冲突，必须承受最小延迟的飞机将执行冲突解决。这架飞机是在另一架飞机后方交叉的飞机，即在最近接近点与另一架飞机的相对方位在 $[-90^\circ, 90^\circ]$ 范围内的飞机。

4. 受限环境中的飞机相互作用

在受限空域中，飞机之间的相互作用与在非受限环境中遇到的相互作用不同。通过分析这些相互作用，可以做出一些假设，这有助于推导平均流入率与飞机密度之间关系的分析模型。这些假设与单个交叉点的最大吞吐量有关，这是分析模型的一个输入。此外，本节中的假设设定了飞机密度的理论上限。

本节介绍了在二维（即所有飞机在同一高度层飞行）受限城市空域环境中可以识别的几种飞机相互作用。为了演示目的，将使用两条单向航路的正交叉点。在这个单一的正交叉点处，区分两种类型的交通。直飞交通代表继续沿直线路径飞行的飞机集合，而转弯交通代表转向另一条航路的飞机集合。首先，评估两条流的直飞交通在单个交叉点处相互作用的影响。接下来，分析

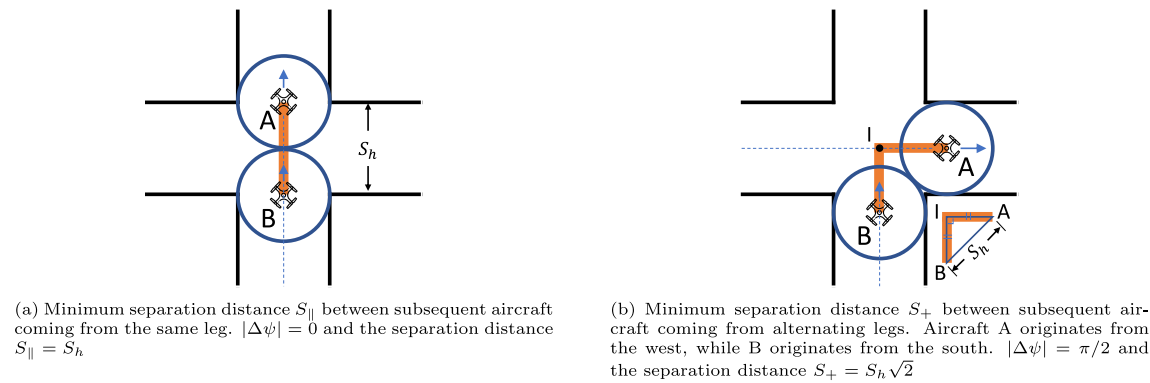


Fig. 2. The minimum separation distance between aircraft on an intersection of two one-way airways, shown visually as the orange bar.

effect of turning traffic interacting with through traffic at this single intersection is analysed. Finally, the resulting macroscopic interactions due to speed-based conflict resolution manoeuvres in an orthogonal grid network environment are examined. The emergent behaviour of the aircraft flow, including queuing for intersections, demonstrates the motivation for a delay model, which will be derived in Section 5.

4.1. Interaction between through traffic of two flows crossing a single intersection

Consider an intersection of two one-way airways, with through traffic only. A difference exists in the minimum separation distance between two aircraft coming from the same upstream leg and between two aircraft coming from different upstream legs. This difference has a significant impact on the maximum throughput capacity of an intersection, as an increased separation margin means that fewer aircraft can cross within the same time window. The increase in minimum separation distance is illustrated in Fig. 2.

In Fig. 2(a), to remain conflict-free, aircraft B must follow aircraft A at a distance of at least the horizontal separation requirement S_h . This safe distance is named the conflict-free flight distance $S_{\angle}$ and is geometrically defined as a function of the absolute difference in ground track angle $|\Delta\psi|$ at the intersection:

$$S_{\angle}(|\Delta\psi|) = \frac{S_h}{\cos(|\Delta\psi|/2)} \quad (2)$$

By using this equation, the minimum separation distance $S_{\parallel}$, shown as the orange bar in Fig. 2(a), between subsequent aircraft coming from the same leg can be determined:

$$S_{\parallel} = S_{\angle}(0) = S_h \quad (3)$$

In the situation where two aircraft originate from different upstream legs, the conflict-free flight distance increases. This situation is visualised in Fig. 2(b), where the orange bar represents the minimum separation distance S_+ for aircraft coming from different upstream legs. In this figure, the conflict-free flight distance is increased to the distance $B \rightarrow I \rightarrow A$, with I signifying the intersection midpoint. For an orthogonal intersection, by applying Eq. (2), S_+ equals:

$$S_+ = S_{\angle}(\pi/2) = S_h\sqrt{2} \quad (4)$$

In summary, after a crossing aircraft of the other airway, the subsequent aircraft needs to keep a separation distance over the path of at least $S_h\sqrt{2}$. However, following a through aircraft of its own airway, a margin of S_h suffices. Therefore, it is assumed that:

Assumption 1. When the origin leg of the crossing traffic at an intersection alternates less often between both upstream legs, intersection throughput capacity can be increased compared to an intersection with perfectly alternating upstream legs.

4.2. Interaction of turning traffic with through traffic on a single intersection

Continuing with the analysis of a single intersection of two one-way airways, turning traffic is added next to the through traffic. Two situations are considered that affect the maximum throughput capacity: either (1) turning aircraft A is in the airway of aircraft B and turning onto the crossing airway, or (2) turning aircraft A is on the crossing airway and merging into the airway of aircraft B. Note that the turns of turning traffic are not predicted by the state-based conflict detection algorithm used in this paper. Additionally, aircraft are assumed to not have to decelerate during a turn.

For the first case, aircraft B can follow aircraft A conflict-free at a distance of $S_{\parallel}$ of Eq. (3) within the airway. As both aircraft have the same airspeed, no conflict is detected. This situation is equivalent to the situation illustrated in Fig. 2(a). However, when aircraft

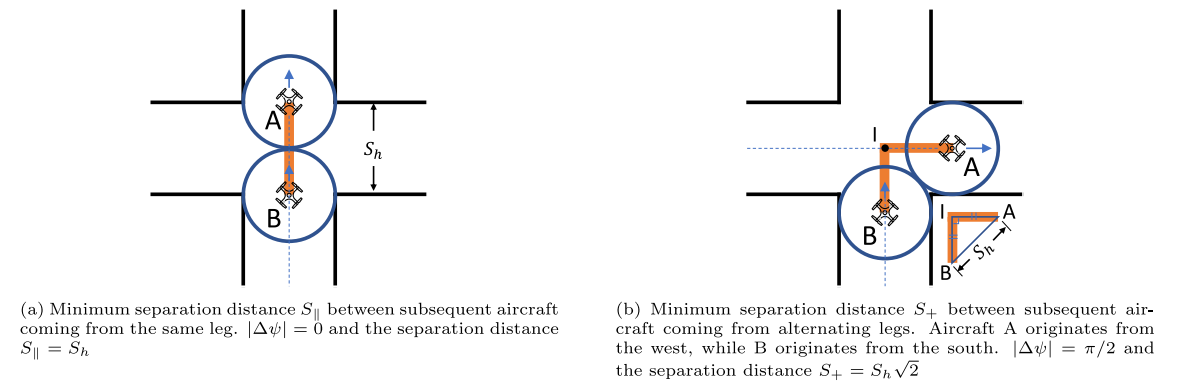


图2. 两条单向航路交叉点处飞机之间的最小间隔距离，直观显示为橙色条。

在这个单一交叉点处转弯交通与直飞交通相互作用的附加影响。最后，研究在正交网格网络环境中基于速度的冲突解决机动导致的宏观相互作用。飞机流的涌现行为，包括在交叉点处排队，展示了延迟模型的动机，将在第5节中推导。

4.1. 两条流的直飞交通在单个交叉点处的相互作用

考虑两条单向航路的交叉点，且仅有直飞交通。来自同一上游航段的两架飞机与来自不同上游航段的两架飞机之间的最小间隔距离存在差异。这种差异对交叉点的最大通行能力有重大影响，因为间隔余量增加意味着在同一时间窗口内能够穿越的飞机数量减少。最小间隔距离的增加情况如图2所示。

在图2(a)中，为保持无冲突，飞机B必须与飞机A保持至少水平间隔要求 S_h 的距离跟随。这个安全距离被称为无冲突飞行距离 $S_{\angle}$ ，并在几何上定义为交叉点处地面航迹角 $|\Delta\psi|$ 的绝对差值的函数：

$$S_{\angle}(|\Delta\psi|) = \frac{S_h}{\cos(|\Delta\psi|/2)} \quad (2)$$

通过使用此方程，可以确定来自同一航段的后续飞机之间的最小间隔距离 $S_{\parallel}$ ，如图2(a)中的橙色条所示：

$$S_{\parallel} = S_{\angle}(0) = S_h \quad (3)$$

在两架飞机来自不同上游航段的情况下，无冲突飞行距离会增加。这种情况在图(b)中可可视化，其中橙色条代表来自不同上游航段的飞机的最小间隔距离 S_+ 。在该图中，无冲突飞行距离增加到距离 $B \rightarrow I \rightarrow A$ ，其中 I 表示交叉点中点。对于正交交叉点，应用式(2)， S_+ 等于：

$$S_+ = S_{\angle}(\pi/2) = S_h\sqrt{2} \quad (4)$$

总之，在另一航路的穿越飞机之后，后续飞机需要在路径上保持至少 $S_h\sqrt{2}$ 的间隔距离。然而，跟随自身航路的直飞飞机时， S_h 的余量就足够了因此，假设：

假设1。当交叉点处交叉交通的起始航段在上游两个航段之间交替的频率较低时，与上游航段完美交替的交叉点相比，交叉点的通行能力可以提高。

4.2. 单交叉口处转弯交通与直飞交通的相互作用

继续分析两条单向航路的单交叉口情况，接下来在直飞交通旁边增加转弯交通。考虑两种影响最大通行能力的情况：（1）转弯飞机A在飞机B的航路上并转向交叉航路，或者（2）转弯飞机A在交叉航路上并汇入飞机B的航路。请注意，本文使用的基于状态的冲突检测算法未预测转弯交通的转向。此外，假设飞机在转弯过程中无需减速。

对于第一种情况，在航路内，飞机B可以在式（3）中 $S_{\parallel}$ 的距离处无冲突地跟随飞机A。由于两架飞机空速相同，未检测到冲突。这种情况等同于图2（a）所示的情况。然而，当飞机

A, instead of continuing through, has turned, a similar situation as visualised in Fig. 2(b) occurs (although aircraft A originated from the bottom leg instead of the left leg). As a result, aircraft B in fact needs to keep a margin equivalent to S_+ of Eq. (4). Hence, when the two aircraft are following each other more closely than S_+ within the airway, the leading aircraft A will provoke a conflict with B by turning. Therefore, each aircraft trailing a turning aircraft within a distance S_+ needs to decelerate and increase its distance to S_+ to resolve the conflict. In other words, this case of turning traffic has a decreasing effect on the intersection throughput capacity for aircraft coming from the same upstream leg.

In the second situation, the opposite occurs. Aircraft B coming from the bottom leg approaches the intersection and expects aircraft A coming from the left leg to continue straight, see Fig. 2(b). As a result, aircraft B predicts an intrusion will occur when its minimum separation distance is smaller than S_+ . Yet, when aircraft A, instead of going straight, turns upwards into the airway of aircraft B, the intrusion will only occur when the minimum separation distance is smaller than $S_{||}$. In other words, by turning, aircraft A moves its closest point of approach away from aircraft B and aircraft B can reduce its conflict-free distance from S_+ to $S_{||}$. As such, each aircraft performing a turn has an increasing effect on the intersection throughput capacity for aircraft coming from the alternating upstream leg.

When the flow rates from both upstream legs are approximately identical, the first situation occurs equally often as the second situation. In addition, the magnitude of change in conflict-free distance is equal, with the sign reversed. Hence, both effects counterbalance each other, resulting in the following assumption:

Assumption 2. Through traffic and turning traffic have an equal impact on the intersection throughput capacity when both upstream flow rates are approximately equal.

In this analysis, it is assumed that aircraft do not decelerate to perform a turn. If this is not the case, an additional delay needs to be modelled per turning aircraft, and turning traffic will have a decreasing impact on maximum throughput capacity compared to through traffic. To prevent this, future airspace designs could implement solutions such as separate altitude layers dedicated to turning traffic (Doole et al., 2021), similar to turning lanes in road traffic.

4.3. Macroscopic interactions due to conflict resolution in constrained urban airspace

In constrained urban airspace, three macroscopic properties due to speed-based conflict resolution need to be considered. The first property, an increase in aircraft density, demonstrates the choice for an analytical delay model as the main input in the relationship between the mean inflow rate and the aircraft density. The second property, a queuing effect, defines an upper bound to the stability of a constrained airspace. This is used as an upper bound in the analytical model as well. The third property is a structuring effect, which assesses the impact of clustered aircraft on the analytical model.

Firstly, an increase in density occurs in a speed-based conflict resolution environment, as aircraft resolve conflicts by decelerating, thereby extending their flight time. This increase can be expressed analytically, starting from the (hypothetical) mean number of instantaneous aircraft without conflict resolution $\bar{N}_{i,NR}$. $\bar{N}_{i,NR}$ and the mean inflow rate $\bar{q}_i$ are related through the mean flight duration (which equals the mean flight distance $\bar{D}$ divided by the cruise speed V_{NR}):

$$\bar{N}_{i,NR} = \frac{\bar{D}}{V_{NR}} \bar{q}_i \quad (5)$$

Aircraft routes and departure times are not impacted by the speed-based conflict resolution used in this paper. Therefore, in a stable situation, an equal number of aircraft depart, arrive or pass an intersection per unit of time, regardless of whether or not conflict resolution is used. Hence, the mean inflow rate $\bar{q}_i$ is unchanged.

The mean airspeed with conflict resolution V_{WR} can be extracted using the increased mean flight duration, which is the sum of the no-resolution flight duration and the mean delay $\bar{d}$:

$$V_{WR} = \frac{\bar{D}}{\frac{\bar{D}}{V_{NR}} + \bar{d}} \quad (6)$$

Subsequently, the actual density, expressed as the mean number of instantaneous aircraft with resolution $\bar{N}_{i,WR}$, is given by substituting V_{NR} with V_{WR} of Eq. (6) into Eq. (5):

$$\bar{N}_{i,WR} = \frac{\bar{D}}{V_{WR}} \bar{q}_i = \left(\frac{\bar{D}}{V_{NR}} + \bar{d} \right) \bar{q}_i \quad (7)$$

The mean route length $\bar{D}$, the cruise speed V_{NR} , and the mean inflow rate $\bar{q}_i$ are assumed to be known input variables. Thus, the mean delay $\bar{d}$ is the only unknown variable required to obtain the mean density and, thereby, to express the entire relationship between the mean flow rate and the mean density. This will be the starting point in the model derivation in the next section.

Secondly, within airways, a queuing effect can occur. While one aircraft waits for a crossing aircraft, another aircraft coming from upstream in the same airway approaches the waiting aircraft and forms a queue, increasing the density locally. When a queue gets longer, it can start to interfere with upstream intersections. This interference is visualised in Fig. 3. In this figure, intersection A is operating above its capacity limit. As a result, not all aircraft can cross the intersection, blocking the aircraft in the northbound leg, with the queue overflowing to the upstream intersection B. In turn, this queue blocks all traffic in the westbound airway upstream of intersection B.

A不继续直飞而是转弯时，会出现类似于图2（b）所示的情况（尽管飞机A来自底部航路而非左侧航路）。结果，飞机B实际上需要保持相当于式（4）中 S_+ 的余量。因此，当两架飞机在航路内彼此跟随的距离小于 S_+ 时，领先的飞机A转弯会与B引发冲突。因此，在距离转弯飞机 S_+ 内跟随的每架飞机都需要减速并将其与转弯飞机的距离增加到 S_+ 以解决冲突。换句话说，这种转弯交通情况对来自同一上游航路的飞机的交叉口通行能力有降低作用。

在第二种情况下，情况相反。来自底部航路的飞机B接近交叉口，并期望来自左侧航路的飞机A继续直飞，见图2（b）。结果，当飞机B的最小间隔距离小于 S_+ 时，它预测会发生侵入。然而，当飞机A不直飞而是向上转入飞机B的航路时，只有当最小间隔距离小于 $S_{||}$ 时才会发生侵入。换句话说，通过转弯，飞机A将其最接近点移离飞机B，并且飞机B可以将其无冲突距离从 S_+ 减小到 $S_{||}$ 。因此，进行转弯的每架飞机对来自交替上游航路的飞机的交叉口通行能力有增加作用。

当来自两个上游航路的流量大致相同时，第一种情况和第二种情况出现的频率相同。此外，无冲突距离的变化幅度相等，只是符号相反。因此，两种影响相互抵消，得出以下假设：

假设2. 当两个上游流量大致相等时，直飞交通和转弯交通对交叉口通行能力的影响相同。

在本分析中，假设飞机转弯时不减速。如果不是这种情况，则需要为每架转弯飞机额外建模延迟，并且与直飞交通相比，转弯交通对最大通行能力的影响将降低。为防止这种情况，未来的空域设计可以实施诸如为转弯交通设置单独高度层（Doole等人，2021年）等解决方案，类似于道路交通中的转弯车道。

4.3. 受限城市空域中冲突解决导致的宏观相互作用

在受限城市空域中，需要考虑基于速度的冲突解决产生的三个宏观特性。第一个特性，飞机密度增加，表明选择分析延迟模型作为平均流入率与飞机密度之间关系的主要输入。第二个特性，排队效应，定义了受限空域稳定性的上限。这在分析模型中也用作上限。第三个特性是结构化效应，它评估聚集飞机对分析模型的影响。

首先，在基于速度的冲突解决环境中会出现密度增加，因为飞机通过减速来解决冲突，从而延长其飞行时间。这种增加可以通过分析来表达，从（假设的）无冲突解决时的瞬时飞机平均数 $\bar{N}_{i,NR} \cdot \bar{N}_{i,NR}$ 开始，平均流入率 $\bar{q}_i$ 通过平均飞行持续时间（等于平均飞行距离 $\bar{D}$ 除以巡航速度 V_{NR} ）相关联：

$$\bar{N}_{i,NR} = \frac{\bar{D}}{V_{NR}} \bar{q}_i \quad (5)$$

本文中基于速度的冲突解决方法不影响飞机航线和起飞时间。因此，在稳定状态下，无论是否采用冲突解决方法，单位时间内通过某一交叉口的飞机数量相等。平均流入率 $\bar{q}_i$ 保持不变。

通过增加的平均飞行持续时间可以提取出采用冲突解决方法时的平均空速 V_{WR} ，平均飞行持续时间是未采用冲突解决方法时的飞行持续时间与平均延误 $\bar{d}$ 之和：

$$V_{WR} = \frac{\bar{D}}{\frac{\bar{D}}{V_{NR}} + \bar{d}} \quad (6)$$

随后，将式（6）中 V_{NR} 替换为 V_{WR} 后代入式（5），得出以具有分辨率 $\bar{N}_{i,WR}$ 的瞬时飞机平均数表示的实际密度：

$$\bar{N}_{i,WR} = \frac{\bar{D}}{V_{WR}} \bar{q}_i = \frac{\bar{D}}{V_{NR}} + \bar{d} \quad \bar{q}_i \quad (7)$$

假定平均航线长度 $\bar{D}$ 、巡航速度 V_{NR} 和平均流入率 $\bar{q}_i$ 为已知输入变量。因此，平均延误 $\bar{d}$ 是获取平均密度从而表示平均流量与平均密度之间完整关系所需的唯一未知变量。这将是下一节模型推导的起点。

其次，在航路上可能会出现排队效应。当一架飞机等待穿越的飞机时，同一航路上从上游飞来的另一架飞机会靠近等待的飞机并形成队列，导致局部密度增加。当队列变长时，可能会开始干扰上游交叉口。图3展示了这种干扰情况。在该图中，交叉口A的运行超出了其容量限制。结果，并非所有飞机都能通过交叉口，阻塞了北向航段中的飞机，队列向上游交叉口B溢出。进而，这个队列阻塞了交叉口B上游西向航路上的所有交通。

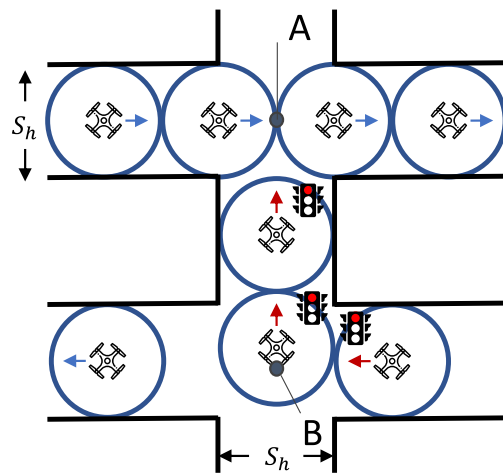


Fig. 3. A small grid network experiencing overflow. Intersection A is operating above its capacity limit, leading to overflow blocking upstream intersection B.

In other words, queuing has a destabilising effect on the network flows, as more and more intersections become occupied by the queuing aircraft, blocking all flows at those intersections. Finally, the entire airspace becomes ‘gridlocked’, as the speed of all aircraft equals 0 m/s and the aircraft are unable to reach their destination. Therefore, it is assumed that once a queue reaches an upstream intersection, the airspace becomes unstable:

Assumption 3. Once the mean queue length within a leg of an intersection is longer than the distance to the upstream intersection of that leg, the airspace becomes unstable.

Due to stochastic effects, the queue length varies and is longer than the mean length at particular intervals. As a result, the network can already become unstable at a lower inflow rate than at the inflow rate needed for the mean length of **Assumption 3**. The assumption is, therefore, expected to overestimate the instability density. Nevertheless, **Assumption 3** is an effective metric to use as a theoretical maximum capacity limit for a given constrained urban airspace, similar to the stability metric defined by Sunil et al. (2018a) for general airspace.

Thirdly, a structuring effect is present. While an aircraft is waiting for a crossing aircraft, a queue forms behind this aircraft. Once the intersection becomes free to cross, the queued aircraft continue their flight bunched together. In other words, aircraft get structured into *platoons* of multiple aircraft following each other closely. These platoons can impact the mean delay incurred at downstream intersections. Namely, a platoon of multiple aircraft can either fit in a relatively tinier gap, decreasing the mean delay for the aircraft in the platoon, or be blocked entirely, increasing the mean delay for the platoon. Furthermore, in this paper, traffic at the intersection adheres to the first-come, first-serve principle because no systems are present to limit flow alternation. Therefore, delay periods are limited to a single crossing aircraft, reducing the size of platoons. Altogether, the effect is assumed negligible for all densities:

Assumption 4. The structuring effect has a negligible impact on the mean delay for all airspace densities.

5. Analytic computations for a constrained urban airspace

We will now first derive an analytical delay model for a single intersection of a two-dimensional orthogonal grid network. This model is based on an analytical delay model for road transportation research and subsequently adjusted to constrained urban airspace. Secondly, combining this delay model and Eq. (7), we approximate the relationship between the mean inflow rate and the mean density in constrained urban airspace. This yields an expression similar to the free-flow part of the Macroscopic Fundamental Diagram. Thirdly, we will comment on airspace stability based on queue lengths, which are used to approximate the transition point of the MFD.

For the models in this paper, we assume that a speed-based conflict resolution algorithm is used, which resolves conflicts by decelerating for a particular period. This way, each conflict results in a delay incurred by a single aircraft.

As a known input variable, we use the mean inflow rate of aircraft in the network $\bar{q}_i$. Together with a known routing distribution and an average route length, this variable can be translated to a mean flow rate in an intersection leg $\bar{q}_f$:

$$\bar{q}_f = \bar{q}_i \frac{\bar{D}}{D_f} p_f \quad (8)$$

In this equation, p_f represents the proportion of aircraft passages through this leg over the sum of all leg passages. D_f is the length of the leg, and $\bar{D}$ the mean flight distance.

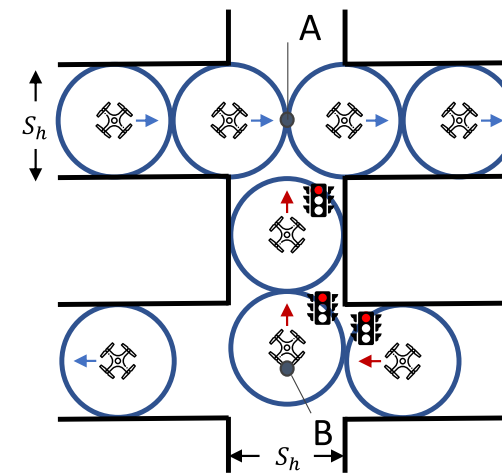


图3. 一个出现溢出的小型网格网络。交叉口A的运行超出其容量限制，导致溢出阻塞上游交叉口B。

换句话说，排队对网络流量具有破坏稳定的作用，因为越来越多的交叉口被排队的飞机占据，阻塞了这些交叉口的所有流量。最终，整个空域会陷入“拥堵”，因为所有飞机的速度都等于 0 m/s，且无法到达目的地。因此，假定一旦队列到达上游交叉口，空域就会变得不稳定：

假设3. 一旦交叉口某一航段内的平均队列长度超过该航段到上游交叉口的距离，空域就会变得不稳定。

由于随机效应，队列长度会发生变化，并且在特定时间段会长于平均长度。因此，网络在低于假设3中平均长度所需流入率的情况下就可能变得不稳定。所以，该假设预计会高估不稳定密度。尽管如此，假设3是一个有效的度量标准，可作为给定受限城市空域的理论最大容量限制，类似于Sunil等人(2018a)为一般空域定义的稳定度量标准。

第三，存在一种编队效应。当一架飞机等待穿越的飞机时，这架飞机后面会形成一个队列。一旦交叉口可以自由穿越，排队的飞机会紧密相随继续飞行。也就是说，飞机会形成由多架飞机紧密跟随的编队。这些编队会影响下游交叉口的平均延误。具体而言，一个由多架飞机组成的编队要么能够适应相对更小的间隙，减少编队中飞机的平均延误，要么会被完全阻塞，增加编队的平均延误。此外，在本文中，交叉口的交通遵循先到先服务原则，因为不存在限制流量交替的系统。因此，延误时间仅限于单个穿越飞机，从而减小了编队规模。总体而言，假定对于所有密度，该效应均可忽略不计：

假设4. 编队效应对所有空域密度下的平均延误影响可忽略不计。

5. 受限城市空域的解析计算

我们现在首先推导二维正交网格网络单个交叉路口的分析延误模型。该模型基于道路交通研究中的分析延误模型，并随后针对受限城市空域进行了调整。其次，将此延误模型与公式（7）相结合，我们近似得出受限城市空域中平均流入率与平均密度之间的关系。这得出了一个类似于宏观基本图自由流部分的表达式。第三，我们将基于队列长度对空域稳定性进行评论，队列长度用于近似宏观基本图的过渡点。

对于本文中的模型，我们假设使用基于速度的冲突解决算法，该算法通过在特定时间段内减速来解决冲突。这样，每次冲突都会导致单个飞机产生延误。

作为一个已知输入变量，我们使用网络中飞机的平均流入率 $\bar{q}_i$ 。连同已知的路由分布和平均航线长度，此变量可以转换为交叉路口路段的平均流率 $\bar{q}_f$ ：

$$\bar{q}_f = \bar{q}_i \frac{\bar{D}}{D_f} p_f \quad (8)$$

在此方程中， p_f 表示通过该路段的飞机通行量占所有路段通行量总和的比例。 D_f 是路段长度， $\bar{D}$ 是平均飞行距离。

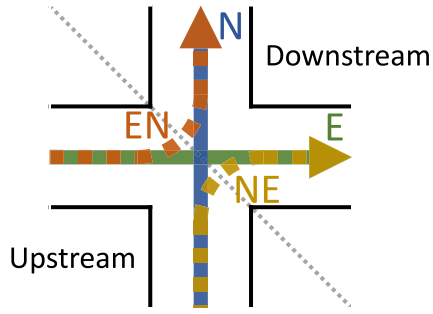


Fig. 4. An orthogonal intersection of area A with two one-way airways and four flows: eastbound (E), northbound (N), east- to northbound turn (EN), and north- to eastbound turn (NE).

5.1. Analytical delay model

Consider the single, same-level, orthogonal intersection with area A of a northbound and an eastbound airway as shown in Fig. 4. Turning onto the other airway is allowed. Therefore, across the intersection, four possible flows exist: an eastbound (E) through flow, a northbound (N) through flow, an east- to northbound (EN) turning flow, and a north- to eastbound (NE) turning flow. These are further grouped into a set of the two upstream legs u , with mean flow rates $\bar{q}_E + \bar{q}_{EN}$ and $\bar{q}_N + \bar{q}_{NE}$, and a set of the two downstream legs d , with mean flow rates $\bar{q}_E + \bar{q}_{NE}$ and $\bar{q}_N + \bar{q}_{EN}$, respectively. The set of all legs F represents the union of the sets of upstream and downstream legs.

5.1.1. Analytical delay in road transportation

Such an intersection has many similarities with signalised intersections found in road transportation. Therefore, as a starting point in the derivation of the delay model, we use Webster's model for delays at signalised intersections (Webster, 1958):

$$\bar{d} = \underbrace{\frac{c(1-\lambda)^2}{2(1-\lambda x)}}_{\text{1st term}} + \underbrace{\frac{x^2}{2\bar{q}(1-x)}}_{\text{2nd term}} - \text{empirical correction factor } (\approx 5\% - 15\%) \quad (9)$$

In this equation, the mean delay incurred in an upstream leg $\bar{d}$ is approximated by two terms, and a correction factor. The first term models the general delay, which is incurred when traffic arrives at a red signal at a constant rate. The second term models the stochastic delay, representing the incurred delay due to the random nature of the separation between consecutive arrivals. A correction factor is included to match the experimental road traffic data found by Webster (1958). In this paper, the empirical correction factor is not used, since there is no experimental data to match. Therefore, the mean delay of aircraft in an upstream leg, $\bar{d}_u$ is given by the sum of the mean general delay $\bar{d}_{g,u}$ and the mean stochastic delay $\bar{d}_{s,u}$:

$$\bar{d}_u = \bar{d}_{g,u} + \bar{d}_{s,u} \quad (10)$$

5.1.2. Analytical delay in constrained urban airspace

To compute the mean density in constrained urban airspace, first, the mean general delay in an upstream leg $\bar{d}_{g,u}$ is approximated. At the intersection of Fig. 4, when assuming that the two flows are perfectly balanced, the maximum number of aircraft that can cross per unit of time is called the flow capacity s of the intersection. This is different per angle of the crossing flow. Therefore, we split the analysis between aircraft crossing at right angles (denoted by subscript $+$) and aircraft crossing after each other originating from the same airway (denoted by subscript $\parallel$). s_+ is the inverse of the time a crossing aircraft needs to clear the intersection for the alternating leg completely, denoted by t_+ . Without other conflicts, t_+ is given by:

$$t_+ = \frac{S_+}{V_{NR}} \quad (11)$$

Hence, for any two crossing flows flying at equal speed, denoted V_{NR} , s_+ can be expressed as:

$$s_+ = \frac{1}{t_+} = \frac{V_{NR}}{S_+ \sqrt{2}} \quad (12)$$

Recall that the separation distance S_+ assumes that the flows are perfectly alternating. When the flows are not perfectly alternating, the mean distance decreases. Therefore, the time to cross t_+ is predicted to be underestimated at regular intersections.

In this paper, no traffic signal is used. Instead, the red signal time r used in the derivation of Webster's model can be replaced by the time t_+ with which a crossing aircraft occupies the intersection. This 'red time' happens at a mean cycle time $\bar{c}_u$ equivalent to the mean time window between two consecutive aircraft within a crossing flow (i.e., the flow originating from the other upstream

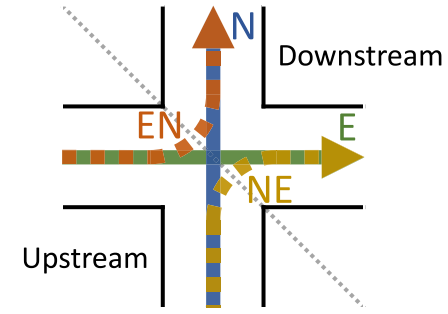


图4. 面积为 A 的正交交叉路口，有两条单向气道和四个流：东向（E）、北向（N）、东转北（EN）和北转东（NE）。

5.1. 分析延误模型

考虑如图4所示的北向和东向气道的单个、同级别、正交交叉路口，其面积为 A 。允许转向另一条气道。因此，在交叉路口处存在四种可能的流：东向（E）直流通行流、北向（N）直流通行流、东转北（EN）转向流和北转东（NE）转向流。这些进一步分为一组两个上游路段 u ，其平均流率分别为 $\bar{q}_E + \bar{q}_{EN}$ 和 $\bar{q}_N + \bar{q}_{NE}$ ，以及一组两个下游路段 d ，其平均流率分别为 $\bar{q}_E + \bar{q}_{NE}$ 和 $\bar{q}_N + \bar{q}_{EN}$ 。所有路段的集合 F 表示上游和下游路段集合的并集。

5.1.1. 道路交通中的分析延误

这样的交叉路口与道路交通中的信号控制交叉路口有许多相似之处。因此，作为延误模型推导的起点，我们使用韦伯斯特模型来计算信号控制交叉路口的延误（韦伯斯特，1958年）：

$$\bar{d} = \underbrace{\frac{c(1-\lambda)^2}{2(1-\lambda x)}}_{\text{1st term}} + \underbrace{\frac{x^2}{2\bar{q}(1-x)}}_{\text{2nd term}} - \text{empirical correction factor } (\approx 5\% - 15\%) \quad (9)$$

在此方程中，上游路段 $\bar{d}$ 产生的平均延误由两项和一个校正因子近似表示。第一项模拟一般延误，即交通以恒定速率到达红灯时产生的延误。第二项模拟随机延误，表示由于连续到达之间间隔的随机性而产生的延误。包含一个校正因子以匹配韦伯斯特（1958年）发现的实验道路交通数据。在本文中，未使用经验校正因子，因为没有可匹配的实验数据。因此，上游路段中飞机的平均延误 $\bar{d}_u$ 由平均一般延误 $\bar{d}_{g,u}$ 和平均随机延误 $\bar{d}_{s,u}$ 之和给出：

$$\bar{d}_u = \bar{d}_{g,u} + \bar{d}_{s,u} \quad (10)$$

5.1.2. 受限城市空域中的分析延误

为了计算受限城市空域中的平均密度，首先，对上游航段 $\bar{d}_{g,u}$ 中的平均总延误进行近似。在图4的交叉路口，假设两个流量完全平衡时，单位时间内能够交叉的最大飞机数量称为交叉路口的流量容量 s 。交叉流的每个角度对应的流量容量都不同。因此，我们将分析分为直角交叉的飞机（用下标 $+$ 表示）和来自同一航路且相继交叉的飞机（用下标 $\parallel$ 表示）。 s_+ 是交叉飞机完全清空交替航段交叉路口所需时间的倒数，用 t_+ 表示。在没有其他冲突的情况下， t_+ 由下式给出：

$$t_+ = \frac{S_+}{V_{NR}} \quad (11)$$

因此，对于以相同速度飞行的任意两个交叉流，用 V_{NR} , s_+ 表示时可表达为：

$$s_+ = \frac{1}{t_+} = \frac{V_{NR}}{S_+ \sqrt{2}} \quad (12)$$

回想一下，分离距离 S_+ 假设流量是完全交替的。当流量不是完全交替时，平均距离会减小。因此，在常规交叉路口，交叉 t_+ 所需的时间预计会被低估。

在本文中，未使用交通信号灯。相反，在韦伯斯特模型推导中使用的红色信号时间 r 可以用交叉飞机占用交叉路口的时间 t_+ 来代替。这个“红色时间”发生在平均周期时间 $\bar{c}_u$ ，它相当于交叉流中两架连续飞机之间的平均时间窗口（即，来自另一个上游

leg), in traffic engineering also known as time headway. This time window is inversely proportional to the mean crossing flow rate $\bar{q}_c$, meaning that:

$$r = t_+ \quad (13)$$

$$\bar{c}_u = 1/\bar{q}_c \quad (14)$$

Hence, the mean proportion of time $\bar{\lambda}$ that the traffic in an upstream leg u perceives the intersection to be free to cross, equals:

$$\bar{\lambda}_u = 1 - \frac{r}{\bar{c}_u} = 1 - t_+ \bar{q}_c \quad (15)$$

When an intersection is free to cross, two aircraft coming from the same upstream leg can cross with a minimum separation time window equal to $t_{\parallel}$:

$$t_{\parallel} = \frac{S_{\parallel}}{V_{NR}} \quad (16)$$

Accordingly, the maximum airway flow capacity $s_{\parallel}$ is inversely proportional to $t_{\parallel}$:

$$s_{\parallel} = \frac{1}{t_{\parallel}} = \frac{V_{NR}}{S_h} \quad (17)$$

Lastly, the mean ratio of the actual flow to the maximum flow which can pass through an intersection from the upstream leg is called the degree of saturation $\bar{x}_u$ and is given by (Webster, 1958):

$$\bar{x}_u = \frac{\bar{q}_u}{\lambda_u s_{\parallel}} \quad (18)$$

We can now obtain an expression for the mean general delay incurred in an upstream leg $\bar{d}_{g,u}$, by substituting all variables defined above into the first part of Eq. (9):

$$\bar{d}_{g,u} = \frac{\bar{c}_u(1 - \bar{\lambda})^2}{2(1 - \bar{\lambda}\bar{x})} = \frac{(1 - t_+ \bar{q}_c)^2}{2\bar{q}_c(1 - t_{\parallel}\bar{q}_u)} \quad (19)$$

Next, the mean stochastic delay in an upstream leg $\bar{d}_{s,u}$ is derived. It concerns the flow within an airway, as it models the delay incurred due to queuing for a waiting aircraft. By inserting Eq. (18) into the second part of Eq. (9), the single intersection airspace version of $\bar{d}_{s,u}$ is obtained:

$$\bar{d}_{s,u} = \frac{\bar{x}_u^2}{2\bar{q}_u(1 - \bar{x}_u)} \quad (20)$$

With expressions for both the mean general delay (Eq. (19)) and the mean stochastic delay (Eq. (20)), the mean delay incurred by an aircraft in an upstream leg can be determined using Eq. (10). As aircraft in the downstream legs d do not encounter any conflicts, no delay is incurred after crossing. Hence, the final set of delay equations is:

$$\bar{d}_u = \frac{(1 - t_+ \bar{q}_c)^2}{2\bar{q}_c(1 - t_{\parallel}\bar{q}_u)} + \frac{\bar{x}_u^2}{2\bar{q}_u(1 - \bar{x}_u)} \quad (21)$$

$$\bar{d}_d = 0 \quad (22)$$

5.2. Analytical free-flow model

The second goal is to get to the relationship between the mean inflow rate $\bar{q}_i$ and the mean density $\bar{N}_{i,WR}$ for the entire airspace. This requires completing Eq. (7). As mentioned, the mean delay $\bar{d}$ was the only unknown variable. In the previous section, we derived an equation for the delay in a single intersection leg. Therefore, we can apply Eq. (7) on a single leg and obtain the mean density in this leg. Recall that using Eq. (8), we could transform the mean inflow rate $\bar{q}_i$ into the flow rate in a single leg $\bar{q}_f$.

By applying Eq. (7) to each leg f of a single intersection, the mean number of instantaneous aircraft with resolution at this intersection $\bar{N}_{i,WR,+}$ can be derived to be:

$$\bar{N}_{i,WR,+} = \sum_{f \in F} \bar{N}_{i,WR,f} = \sum_{f \in F} \left(\frac{D_f}{V_{NR}} + \bar{d}_f \right) \bar{q}_f \quad (23)$$

Under Assumption 2, Eqs. (21) to (23) are valid for intersections with and without turning traffic. Also, all interactions between intersections are either assumed negligible (Assumption 4) or limiting for the entire airspace (Assumption 3). Hence, the mean number of instantaneous aircraft with resolution in the grid network can be obtained by summing $\bar{N}_{i,WR}$ per leg z for the set of all

航段的流量), 在交通工程中也称为车头时距。这个时间窗口与平均交叉流率 $\bar{q}_c$ 成反比, 即:

$$r = t_+ \quad (13)$$

$$\bar{c}_u = 1/\bar{q}_c \quad (14)$$

因此, 上游航段 u 中的交通流认为交叉路口可自由交叉的平均时间比例 $\bar{\lambda}$ 等于:

$$\bar{\lambda}_u = 1 - \frac{r}{\bar{c}_u} = 1 - t_+ \bar{q}_c \quad (15)$$

当交叉路口可自由交叉时, 来自同一上游航段的两架飞机可以以等于 $t_{\parallel}$ 的最小分离时间窗口交叉:

$$t_{\parallel} = \frac{S_{\parallel}}{V_{NR}} \quad (16)$$

相应地, 最大航路流量容量 $s_{\parallel}$ 与 $t_{\parallel}$ 成反比:

$$s_{\parallel} = \frac{1}{t_{\parallel}} = \frac{V_{NR}}{S_h} \quad (17)$$

最后, 实际流量与从上游航段通过交叉路口所能通过的最大流量的平均比率称为饱和度 $\bar{x}_u$, 由下式给出 (韦伯斯特, 1958年):

$$\bar{x}_u = \frac{\bar{q}_u}{\lambda_u s_{\parallel}} \quad (18)$$

通过将上述定义的所有变量代入式 (9) 的第一部分, 我们现在可以得到上游航段 $\bar{d}_{g,u}$ 中产生的平均总延误的表达式:

$$\bar{d}_{g,u} = \frac{\bar{c}_u(1 - \bar{\lambda})^2}{2(1 - \bar{\lambda}\bar{x})} = \frac{(1 - t_+ \bar{q}_c)^2}{2\bar{q}_c(1 - t_{\parallel}\bar{q}_u)} \quad (19)$$

接下来, 推导上游航段 $\bar{d}_{s,u}$ 中的平均随机延误。它涉及航路内的流量, 因为它模拟了等待飞机排队所产生的延误。通过将式 (18) 代入式 (9) 的第二部分, 可得到 $\bar{d}_{s,u}$ 的单交叉路口空域版本:

$$\bar{d}_{s,u} = \frac{\bar{x}_u^2}{2\bar{q}_u(1 - \bar{x}_u)} \quad (20)$$

有了平均总延误 (式 (19)) 和平均随机延误 (式 (20)) 的表达式, 就可以使用式 (10) 确定飞机在上游航段产生的平均延误。由于下游航段 d 中的飞机不会遇到任何冲突, 交叉后不会产生延误。因此, 最终的延误方程组为:

$$\bar{d}_u = \frac{(1 - t_+ \bar{q}_c)^2}{2\bar{q}_c(1 - t_{\parallel}\bar{q}_u)} + \frac{\bar{x}_u^2}{2\bar{q}_u(1 - \bar{x}_u)} \quad (21)$$

$$\bar{d}_d = 0 \quad (22)$$

5.2. 解析自由流模型

第二个目标是确定空域平均流入率 $\bar{q}_i$ 与平均密度 $\bar{N}_{i,WR}$ 的关系, 需完善式 (7)。平均延误 $\bar{d}$ 是唯一未知变量。上一节已推导单交叉路口航段延误方程, 可将式 (7) 应用于单个航段得到其平均密度。使用式 (8), 可将平均流入率 $\bar{q}_i$ 转换为单个航段流量 $\bar{q}_f$ 。

通过将式 (7) 应用于单个交叉路口的每个航段 f , 可得出该交叉路口具有冲突解决方案时的瞬时飞机平均数量 $\bar{N}_{i,WR,+}$ 为:

$$\bar{N}_{i,WR,+} = \sum_{f \in F} \bar{N}_{i,WR,f} = \sum_{f \in F} \left(\frac{D_f}{V_{NR}} + \bar{d}_f \right) \bar{q}_f \quad (23)$$

在假设2下, 式 (21) 至式 (23) 对于有转弯交通和无转弯交通的交叉路口均有效。此外, 交叉路口的所有相互作用要么被假定为可忽略不计 (假设4), 要么对整个空域起限制作用 (假设3)。因此, 通过对所有集合中每条边 z 的 $\bar{N}_{i,WR}$ 求和, 可以得到网格网络中具有冲突解决方案的瞬时飞机的平均数量

legs Z in the network:

$$\begin{aligned}\bar{N}_{i,W R,grid} &= \sum_{z \in Z} \bar{N}_{i,W R,z} \\ &= \sum_{z \in Z} \left(\frac{D_l}{V_{NR}} + \bar{d}_z \right) \bar{q}_z \\ &= \bar{q}_i \bar{D} \sum_{z \in Z} \left(\frac{D_z}{V_{NR}} + \bar{d}_z \right) \frac{p_z}{D_z}\end{aligned}\tag{24}$$

By substituting Eqs. (21) and (22) into Eq. (24), the complete relationship between the mean inflow rate and the mean density in the airspace is derived. This relationship is similar to the free-flow part of the Macroscopic Fundamental Diagram used in road transportation research. As such, it can be used by future urban airspace designers to assess the capacity of a constrained urban airspace.

Additionally, following Assumption 1, the derived analytical delay model is assumed to act as an upper bound to the experimental mean delay, due to the overestimation of the time t_+ with which a crossing aircraft occupies an intersection. This overestimation is increased when more aircraft from the same origin leg cross without alternating with traffic from the crossing leg, i.e., for increasing density or flow ratio unbalance.

5.3. Airspace capacity measure

We are interested in the theoretical capacity (i.e., the maximum stable flow) and the factors influencing this. To obtain a measure of the transition point between free-flow (stable) and congestion (unstable), we approximate a measure of stability based on the mean queue length (in meters) $\bar{L}_z$, as stated by Assumption 3. The mean queue length in a leg $\bar{L}_z$ can be calculated using Little's law (Little, 1961) for the number of vehicles, multiplied by the horizontal separation requirement S_h :

$$\bar{L}_z = \bar{d}_z \bar{q}_z S_h\tag{25}$$

By Assumption 3, when $\bar{L}_z$ is greater than the distance to the next intersection, i.e., length of the leg D_z , the airspace is predicted to become unstable.

6. Simulations

To verify the assumptions and models from the previous sections, a set of simulation experiments is performed. All experiments are conducted using the BlueSky open-source ATM simulator (Hoekstra and Ellerbroek, 2016).¹ This is a microscopic simulator representing the movements of individual aircraft. The remainder of this section will first discuss the tactical conflict detection and resolution algorithms used in the simulation. This is followed by a summary of the scenarios generated for the experiment. Finally, a model accuracy parameter is proposed, facilitating the comparison between the analytical model predictions and the simulation results. Results will be presented in Section 7.

6.1. Conflict detection and resolution

The state-based conflict detection algorithm used in this study is equal to the algorithm used by Sunil et al. (2018a) and assumes perfect knowledge of aircraft position and ground speed vector. In the algorithm, aircraft positions are linearly extrapolated along their ground speed vectors over a look-ahead time t_l of 20 seconds to predict intrusions.

Tactical conflict resolution is implemented, where each pairwise conflict is resolved using a speed-based resolution algorithm. This algorithm dictates that one aircraft decelerates to avoid the predicted intrusion. Recall from Eq. (1) that the resolution speed V_{WR} is in the range $[0, V_{NR})$. During a simulation, both the detection and resolution algorithms are executed for each simulation timestep Δt of 0.05 s.

6.2. Scenario generation

We aim to set up a simulation in which we can capture the essence of the elements we want to study. There should hence be interactions between aircraft, and since we consider a constraint air space (e.g., between buildings), we consider a network with a road-like layout in the horizontal plane. In order to get the most insights from the resulting simulation results, we choose the simplest form of this. This way, results are not caused by network artefacts or irregularities. We hence simulate an orthogonal grid, with nodes where two one-way airways intersect, with the outer airways forming a counterclockwise flow, as depicted in Figs. 5 and 6. Each aircraft is randomly assigned an origin (O) and a (different) destination (D) node from the set of OD nodes. This set includes all nodes halfway between two intersections. The grid is set up with a separation of 400m between two adjacent intersections, resulting in a total simulation area of almost 8km². To prevent instantaneous intrusions between aircraft departing subsequently,

网络中的航段 Z

$$\begin{aligned}\bar{N}_{i,W R,grid} &= \sum_{z \in Z} \bar{N}_{i,W R,z} \\ &= \sum_{z \in Z} \left(\frac{D_l}{V_{NR}} + \bar{d}_z \right) \bar{q}_z \\ &= \bar{q}_i \bar{D} \sum_{z \in Z} \left(\frac{D_z}{V_{NR}} + \bar{d}_z \right) \frac{p_z}{D_z}\end{aligned}\tag{24}$$

通过将方程(21)和(22)代入方程(24), 推导出空域中平均流入率与平均密度之间的完整关系。这种关系类似于道路交通研究中使用的宏观基本图的自由流部分。因此, 未来的城市空域设计者可以使用它来评估受限城市空域的容量。

此外, 根据假设1, 由于交叉飞机占用交叉点的时间 t_+ 被高估, 所推导的分析延迟模型被假定为实验平均延迟的上限。当更多来自同一起始航段的飞机不与交叉航段的交通交替而交叉时, 即密度或流量比不平衡增加时, 这种高估会加剧。

5.3. 空域容量度量



我们关注理论通行能力(即最大稳定流量)及其影响因素。为了获得自由流(稳定)和拥堵(不稳定)之间过渡点的度量, 我们根据假设3, 基于平均队列长度(以米为单位) $\bar{L}$ 来近似稳定性度量。一条支路 $\bar{L}_z$ 中的平均队列长度可以使用利特尔法则(利特尔, 1961年)计算车辆数量, 再乘以水平间隔要求 S_h :

$$\bar{L}_z = \bar{d}_z \bar{q}_z S_h\tag{25}$$

根据假设3, 当 $\bar{L}_z$ 大于到下一个交叉点的距离, 即航段 D_z 的长度时, 预计空域将变得不稳定。

6. 模拟

为验证上一节中的假设和模型, 进行了一组仿真实验。所有实验均使用BlueSky开源空中交通管理模拟器(Hoekstra和Ellerbroek, 2016年)进行。¹ 这是一个微观模拟器, 用于表示单个飞机的运动。本节其余部分将首先讨论仿真中使用的战术冲突检测和解决算法。接下来是为实验生成的场景总结。最后, 提出一个模型精度参数, 便于比较分析模型预测结果和仿真结果。结果将在第7节中呈现。

6.1. 冲突检测与解决

本研究中使用的基于状态的冲突检测算法与Sunil等人(2018a)使用的算法相同, 并且假定对飞机位置和地速向量有完全的了解。在该算法中, 飞机位置沿着其地速向量在20秒的前瞻时间 t_l 内进行线性外推, 以预测侵入情况。

实施战术冲突解决, 其中使用基于速度的解决算法解决每对冲突。该算法规定一架飞机减速以避免预测的侵入。从式(1)中回想, 解决速度 V_{WR} 在 $[0, V_{NR})$ 范围内。在模拟期间, 针对 0.05 s 的每个模拟时间步长 Δt 执行检测和解决算法。

6.2. 场景生成

我们旨在建立一个模拟, 以便能够捕捉我们想要研究的元素的本质。因此, 飞机之间应该存在相互作用, 并且由于我们考虑的是受限空域(例如, 建筑物之间), 所以我们考虑在水平面上具有类似道路布局的网络。为了从模拟结果中获得最多的见解, 我们选择其最简单的形式。这样, 结果就不会由网络假象或不规则性导致。因此, 我们模拟一个正交网格, 在两条单向气道相交的节点处, 外部气道形成逆时针流动, 如图5和图6所示。每架飞机从OD节点集合中随机分配一个起点(O)和一个(不同的)终点(D)节点。该集合包括两个交叉点之间的所有中间节点。网格设置为相邻两个交叉点之间的间距为 400 m, 从而得到总面积约为 8 km² 的模拟区域。为了防止后续起飞的飞机之间出现瞬时侵入情况,

¹ The source code used to perform the simulations can be found here: <https://doi.org/10.5281/zenodo.5984628>

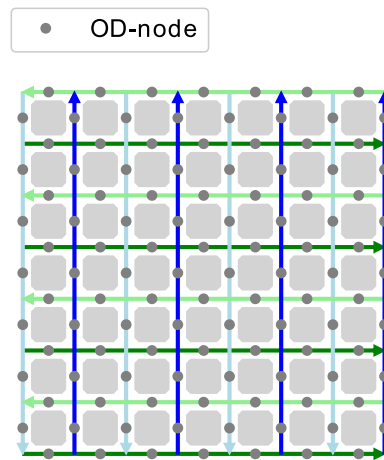
Fig. 5. Orthogonal grid network structure, spanning an area of almost 8 km².

Fig. 6. An example simulation run through a part of the orthogonal grid network in BlueSky, with the path of UAV0000 shown in purple.

the origin nodes of the three prior departures are excluded from the set of origin nodes for that aircraft. However, instantaneous intrusions with cruising aircraft can still occur.

The route is chosen based on the shortest path between the origin and destination node, taking into account the one-way restrictions. When multiple paths are equally short, the path with the least number of turns is chosen. As a result, only a single route is available for any OD combination.

For all conditions, scenarios are generated in which we aim for a more or less constant traffic density, by using a stochastically constant inflow. In contrast to Sunil et al. (2018b), aircraft are introduced at intervals following a random Poisson variable, i.e., exponentially distributed with a mean inflow rate $\bar{q}_i$. Table 1 summarises the common parameters used in the simulations.

We will be running the simulation with 3 different settings. Firstly, a base experiment with the parameters outlined in Table 1 is used to evaluate the accuracy of the analytical free-flow model. Secondly, a speed experiment with a cruise speed of 5 m/s is used as a sensitivity analysis to verify and show the impact of changing V_{NR} . Thirdly, a horizontal separation requirement experiment is used with an S_h of 100 meters to verify and show the impact of changing S_h on the relationship between mean inflow rate and the mean density.

Both the base and speed experiments are run at 10 different traffic inflow rates $\bar{q}_i$ equally spaced from 22 to 220 aircraft per second. The maximum mean traffic density of $220 \bar{N}_{i,NR}$ is chosen as the upper bound, since for higher inflow rates simulation runs became unstable (i.e., the simulated airspace became gridlocked, see Section 4.3 and (Aarts, 2021)). For the separation experiment, the runs were found to become unstable upward of a lower mean traffic density compared to the base and speed experiment, namely from $80 \bar{N}_{i,NR}$. Therefore, as a maximum mean density, 80 is chosen, and the separation experiment uses 10 densities equally spaced

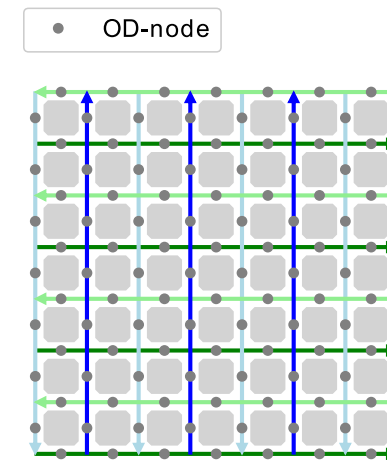
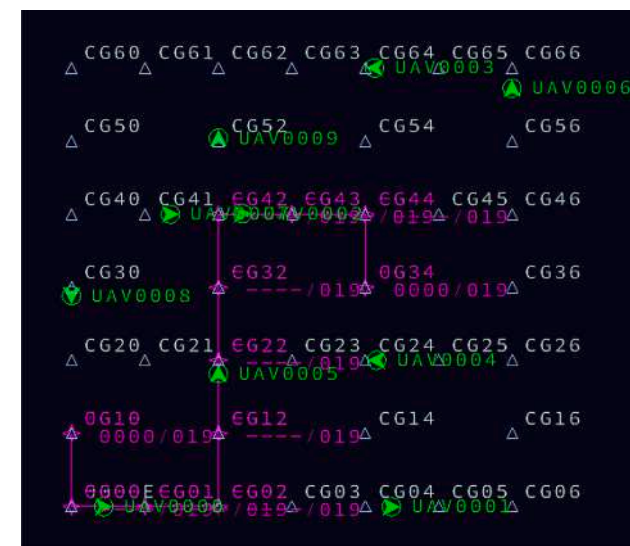
图5. 正交网格网络结构，覆盖面积约为 8 km²。

图6. 在BlueSky 中通过正交网格网络一部分的模拟运行示例，无人机0000 的路径以紫色显示。

该飞机的起点节点集合中排除了前三架起飞飞机的起点节点。然而，与巡航飞机的瞬时侵入情况仍可能发生。

路线根据起点和终点节点之间的最短路径选择，同时考虑单向限制。当有多条路径同样短时，选择转弯次数最少的路径。因此，对于任何OD组合，只有一条路线可用。

通过随机恒定流入量生成场景，使交通密度基本保持稳定。与Sunil等人（2018b）不同，飞机按随机泊松变量间隔引入，即按平均流入率 $\bar{q}_i$ 呈指数分布。表1总结了模拟参数。

我们将使用3种不同设置运行模拟。首先，使用表1中概述的参数进行基础实验，以评估解析自由流模型的准确性。其次，进行巡航速度为 5 m/s 的速度实验，作为敏感性分析，以验证并展示改变 V_{NR} 的影响。第三，进行水平间隔要求实验，水平间隔 S_h 为100米，以验证并展示改变 S_h 对平均流入率与平均密度之间关系的影响。

基础实验和速度实验均在每秒22至220架飞机等间距分布的10种不同交通流入率 $\bar{q}_i$ 下运行。选择最大平均交通密度 $220 \bar{N}_{i,NR}$ 作为上限，因为对于更高的流入率，模拟运行会变得不稳定（即模拟空域会出现拥堵，见第4.3节和（Aarts, 2021））。对于间隔实验，与基础实验和速度实验相比，发现运行在较低的平均交通密度之上就会变得不稳定，即从 $80 \bar{N}_{i,NR}$ 开始。因此，选择最大平均密度为80，间隔实验使用10个等间距的密度

Table 1
Parameters used in the simulations.

Parameter	Value	Unit	Description
a	3.5	m/s ²	Acceleration or deceleration capability
D	200	m	Distance OD-node to intersection
S_h	50	m	Horizontal separation requirement
t_l	20	s	Look-ahead time
V_{NR}	10	m/s	Cruise speed of aircraft NR
V_{WR}	$[0, V_{NR})$	m/s	Possible range of speeds of aircraft WR (see Eq. (1))
Δt	0.05	s	Simulation timestep
Repetitions	10	–	Number of repetitions per density for each experiment
Size	8x8	–	Number of flows (H x V)

Table 2
Summary of experiment inputs.

Experiment	Changed variable	$\overline{N}_{i,NR}$ range	Total no. of aircraft ['000]
Base	–	[22, 44, ..., 220]	377
$0.5V_{NR}$	$V_{NR} \rightarrow 5$ m/s	[22, 44, ..., 220]	188
$2S_h$	$S_h \rightarrow 100$ m	[8, 16, ..., 80]	137

from 8 to $80\overline{N}_{i,NR}$. The inputs of the experiments are summarised in Table 2. 10 different random realisations are generated for each traffic density, to account for random fluctuations. Hence, altogether, 300 scenarios are simulated.

In each scenario, all aircraft are assigned a departure time, origin, destination, and route. Each run uses a warm-up period of 15 minutes, a logging period of 45 minutes, and a cool-down period of 15 minutes in which aircraft keep spawning. The warm-up period is used to avoid effects at the start of the simulation, due to the generation of aircraft. The cool-down period, likewise, is used to avoid effects at the end, due to aircraft arriving. 15 minutes is deemed sufficient, as the maximum trip duration without resolution is less than 6 minutes in all simulation runs. Furthermore, when evaluating dependent airspace variables, such as the mean density or the mean delay, the values over the entire logging period are averaged.

6.3. Model accuracy assessment

A qualitative assessment can be made of the accuracy by inspecting the similarity in shape and magnitude between the analytical model predictions and the simulation results. To also facilitate a quantitative comparison, the k -parameter is introduced (Sunil et al., 2018a):

$$q_{\text{Simulation measurement}} = q_{\text{Analytical model}} \cdot k \quad (26)$$

This parameter is obtained by fitting the analytical model in a least-squares sense to the simulation data. In this report, we convert k to express the relative error, $k_{\%}$, as well:

$$k_{\%} = \left(1 - \left|\frac{k-1}{k}\right|\right) \cdot 100\% \quad (27)$$

We expect that not all simulations will produce stable simulation results. For the same inflow, we repeat the simulation several times with stochastic changes. The relative frequency at which simulations for a certain inflow will become unstable is an important criterion to assess how close the inflow value is to the transition point between free flow and congestion, and, thereby, the capacity of the network.

7. Results

This section compares the predictions of the analytical free-flow model to the outcomes of the simulations. Next, the section analyses the correlation between the analytical model’s prediction of the theoretical capacity of the constrained airspace and the maximum capacities encountered in the simulation.

7.1. Analytical free-flow model validation

In Fig. 7, the comparison between the predictions of the analytical model and the results of the simulations is visualised. The analytical model predictions of Section 5.2 are shown as solid lines, with the corresponding simulation runs marked by an ‘x’ in the same colour. As can be seen, the analytical model closely predicts both the shape and the magnitude of the mean density. Only for higher inflow rates the analytical model slightly overestimates the simulated mean densities.

Simulation runs with the highest densities sometimes became unstable, i.e., 8, 2, and 5 out of the 10 random realisations for the maximum density simulation runs of the base, $0.5V_{NR}$, and $2S_h$ experiments, respectively. In an unstable run, the airspace became gridlocked. In other words, the speed of all aircraft had reduced to 0 m/s and they were unable to reach their destination. These

表1
模拟中使用的参数。

Parameter	Value	Unit	Description
a	3.5	m/s ²	Acceleration or deceleration capability
D	200	m	Distance OD-node to intersection
S_h	50	m	Horizontal separation requirement
t_l	20	s	Look-ahead time
V_{NR}	10	m/s	Cruise speed of aircraft NR
V_{WR}	$[0, V_{NR})$	m/s	Possible range of speeds of aircraft WR (see Eq. (1))
Δt	0.05	s	Simulation timestep
Repetitions	10	–	Number of repetitions per density for each experiment
Size	8x8	–	Number of flows (H x V)

表2
实验输入总结。

Experiment	Changed variable	$\overline{N}_{i,NR}$ range	Total no. of aircraft ['000]
Base	–	[22, 44, ..., 220]	377
$0.5V_{NR}$	$V_{NR} \rightarrow 5$ m/s	[22, 44, ..., 220]	188
$2S_h$	$S_h \rightarrow 100$ m	[8, 16, ..., 80]	137

从 $80\overline{N}_{i,NR}$ 到8。为每个交通密度生成10种不同的随机实现，以考虑随机波动。因此，总共模拟300个场景。

在每种情况下，都会为所有飞机分配出发时间、起点、目的地和航线。每次运行都使用15分钟的预热期、45分钟的记录期和15分钟的冷却期，在此期间飞机持续生成。预热期用于避免由于飞机生成而在模拟开始时产生的影响。同样，冷却期用于避免由于飞机到达而在结束时产生的影响。15分钟被认为是足够的，因为在所有模拟运行中，无冲突解决情况下的最大行程持续时间小于6分钟。此外，在评估相关空域变量（如平均密度或平均延迟）时，会对整个记录期内的值进行平均。

6.3. 模型准确性评估

通过检查分析模型预测与模拟结果之间在形状和大小上的相似性，可以对准确性进行定性评估。为了便于进行定量比较，引入了 k 参数（Sunil等人，2018a）：

$$q_{\text{Simulation measurement}} = q_{\text{Analytical model}} \cdot k \quad (26)$$

该参数通过将分析模型以最小二乘法拟合到模拟数据中获得。在本报告中，我们还将 k 转换为相对误差 $k_{\%}$ 来表示：

$$k_{\%} = 1 - \left|\frac{k-1}{k}\right| \cdot 100\% \quad (27)$$

我们预计并非所有模拟都会产生稳定的模拟结果。对于相同的流入量，我们对模拟进行多次随机变化的重复。对于特定流入量的模拟变得不稳定的相对频率是评估流入量值与自由流和拥堵之间过渡点的接近程度的重要标准，从而也是评估网络容量的重要标准。

7. 结果

本节将分析性自由流模型的预测与模拟结果进行比较。接下来，分析模型对受限空域理论容量的预测与模拟中遇到的最大容量之间的相关性。

7.1. 分析性自由流模型验证

在图7中，展示了分析模型预测与模拟结果之间的比较。第5.2节的分析模型预测以实线表示，相应的模拟运行在图中用相同颜色的“x”标记。可以看出，分析模型能够很好地预测平均密度的形状和大小。仅对于较高的流入率，分析模型略微高估了模拟的平均密度。

密度最高的模拟运行有时会变得不稳定，即分别在基础实验、 $0.5V_{NR}$ 实验和 $2S_h$ 实验的最大密度模拟运行的10次随机实现中，有8次、2次和5次出现这种情况。在不稳定的运行中，空域会陷入拥堵。换句话说，所有飞机的速度都降至 0 m/s，并且无法到达目的地。这些

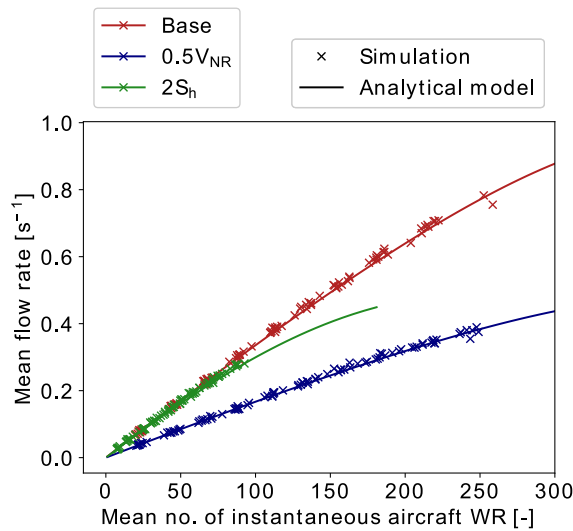


Fig. 7. Comparison between the analytical model (solid line) and the simulation data (x) for the relationship between the mean inflow rate and the mean density for the three orthogonal grid network experiments.

Table 3

Accuracy of the analytical model for the relationship between the mean inflow rate and the mean density compared to the simulation experiments.

Experiment	k	$k_{\%}$	N (unstable at $\bar{q}_{l,max}$)
Base	0.984	98.4%	8/10
$0.5V_{NR}$	0.981	98.1%	2/10
$2S_h$	0.979	97.8%	5/10

unstable runs cannot be shown in the figures, as, without outflow, the number of instantaneous aircraft keeps increasing linearly with inflow.

In addition, the comparison between the analytical model and the simulation experiments is expressed quantitatively in terms of k and $k_{\%}$ (as defined by Eqs. (26) and (27)) in Table 3. Note that the fitted model curve, i.e., the model multiplied by k , is not shown in the figure. The accuracies of 98% and higher show that the model can closely predict the shape and magnitude of the relationship between the mean inflow rate and the mean density in a two-dimensional grid network airspace. Inspecting the results closely, we find all three experiments show a slight overestimation of the mean density predicted for a given mean inflow rate (the line is below the points, or numerically we find values of k in the order of 0.98, which is smaller than 1).

This slight overestimation can be explained by noting that, as mentioned in Section 5, the derived model acts as an upper bound on delay and, therefore, as an upper bound on the mean density. Namely, the analytical model overestimates the time a crossing aircraft occupies an intersection when the upstream legs are not perfectly alternating. In the grid network airspace used in the simulations, the busy intersections near the centre are unbalanced, having flow ratios between both upstream legs up to approximately 1.7 : 1. This imbalance explains the overestimation of the mean delay and the mean density.

7.2. Comparison between theoretical and simulation airspace capacities

The analytical model allows the prediction of the theoretical maximum density and flow rate for an airspace with given parameters (e.g., cruise speed, size, separation distance), using Assumption 3 and Eq. (25). The calculated values for the three experiments are shown under ‘Analytical’ in Table 4. Next to the analytical predictions, the maximum densities as encountered in the simulation runs are shown under ‘Simulated’. Due to randomness in the simulation runs, no hard upper bound for the maximum capacity can be derived. However, all three experiments produced at least a single unstable run at their maximum density (see Table 3). Therefore, the maximum capacities encountered in the simulations are assumed to be a close approximation to the actual maximum.

The table shows that a change in cruise speed V_{NR} does not impact the analytically predicted maximum density. Furthermore, the same density can be reached by scaling the network inflow rate proportionally to the cruise speed. However, for a change in the horizontal separation requirement S_h , no proportionality is found. Increasing S_h means that both the maximum mean density and the maximum mean flow rate decrease. These analytical predictions are confirmed by the simulation data, which, although at a lower maximum capacity, shows similar behaviour.

Additionally, the maximum theoretical densities as predicted by the analytical model overestimate the maximum stable densities encountered in the simulation runs by a factor of 1.35 for the Base and $0.5V_{NR}$ experiments and a factor of almost 2 for the $2S_h$

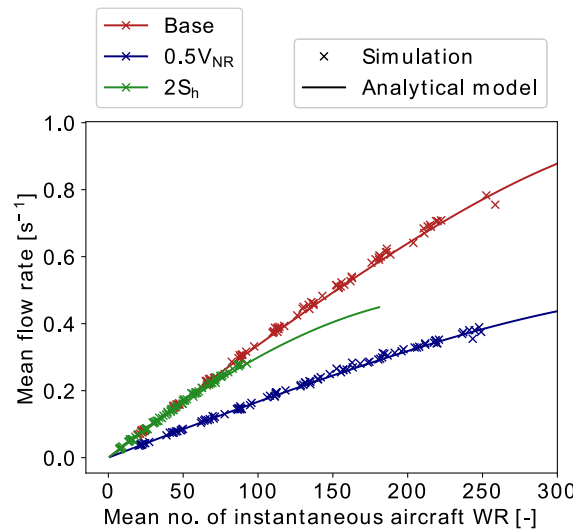


图7. 三个正交网格网络实验中平均流入率与平均密度关系的分析模型(实线)与模拟数据(x)比较。

表3

与模拟实验相比，分析模型对平均流入率与平均密度关系的准确性。

Experiment	k	$k_{\%}$	N unstable at $\bar{q}_{l,max}$
Base	0.984	98.4%	8/10
$0.5V_{NR}$	0.981	98.1%	2/10
$2S_h$	0.979	97.8%	5/10

不稳定的运行无法在图中显示，因为在没有流出的情况下，瞬时飞机数量会随着流入量线性增加。

此外，分析模型与模拟实验之间的比较在表3中以 k 和 $k_{\%}$ （如式（26）和（27）所定义）进行定量表示。请注意，拟合模型曲线（即乘以 k 的模型）未在图中显示。98% 及更高的准确性表明该模型能够很好地预测二维网格网络空域中平均流入率与平均密度关系的形状和大小。仔细检查结果，我们发现所有三个实验都显示出对于给定平均流入率预测的平均密度略有高估（线在点下方，或者从数值上我们发现 k 的均值约为0.98，小于1）。

这种轻微的高估可以通过以下方式解释：正如第5节中所提到的，推导的模型作为延迟的上限，因此也是平均密度的上限。也就是说，当上游路段不完全交替时，分析模型高估了交叉飞机占用交叉点的时间。在模拟中使用的网格网络空域中，中心附近繁忙的交叉点是不平衡的，两个上游路段之间的流量比高达约 1.7:1。这种不平衡解释了平均延迟和平均密度的高估。

7.2. 理论空域容量与模拟空域容量的比较

分析模型可依据假设3和公式(25)，预测给定参数（如巡航速度、尺寸、间隔距离）的空域的理论最大密度和流量。三个实验的计算值列于表4的“分析”一栏。除了分析预测值外，模拟运行中遇到的最大密度列于“模拟”一栏。由于模拟运行存在随机性，无法得出最大容量的确切上限。然而，所有三个实验在其最大密度时都至少产生了一次不稳定运行（见表3）。因此，模拟中遇到的最大容量被假定为与实际最大值非常接近。

该表显示，巡航速度 V_{NR} 的变化不会影响分析预测的最大密度。此外，通过按比例调整网络流入速率与巡航速度，可以达到相同的密度。然而，对于水平间隔要求 S_h 的变化，未发现比例关系。增加 S_h 意味着最大平均密度和最大平均流量都会降低。这些分析预测得到了模拟数据的证实，模拟数据虽然最大容量较低，但显示出类似的行为。

此外，分析模型预测的最大理论密度比基础实验和 $0.5V_{NR}$ 实验模拟运行中遇到的最大稳定密度高估了1.35倍，比 $2S_h$ 实验高估了近2倍

Table 4
Analytical versus simulated capacities of the three experiments, shown as a combination of maximum stable density and stable inflow rate.

Experiment	Analytical		Simulated	
	Max. $\overline{N}_{i,W,R}$	Max. $\overline{q}_i$ s ⁻¹	Max. $\overline{N}_{i,W,R}$	Max. $\overline{q}_i$ s ⁻¹
Base	353	0.96	259	0.78
0.5 V_{NR}	353	0.48	249	0.39
2 S_h	181	0.45	93	0.28

experiment. Yet, this was predicted by [Assumption 3](#), as stochastic effects mean that the network can already become unstable at a lower flow rate than at the mean flow rate used for the analytical prediction.

8. Comparison and synthesis

In this study, a purely analytical free-flow model for the relationship between the mean inflow rate and the mean density was developed to investigate and quantify the factors influencing the capacity of constrained urban airspace. Simulations were performed to validate the accuracy of the analytical model. This section discusses the level of accuracy of the analytical model, followed by a discussion on the effects of several airspace design parameters on its capacity.

8.1. Analytical free-flow model validation

The comparison of the analytical model with the results of simulations showed that model accuracy is high, with accuracies higher than 97%. The analytical model closely predicts the shape of the relationship between the mean inflow rate and the mean density, as well as the values. The inaccuracies stem from an overestimation of the mean density by less than 2.5% in all three experiments.

Two limitations were identified in [Section 7](#). The first limitation is the slight overestimation of the mean density. As mentioned, this is the result of the overestimation of the mean delay on unbalanced intersections. The second limitation is that the maximum theoretical densities as predicted by the analytical model overestimate the maximum stable densities encountered in the simulation runs. This can be attributed to the stochastic periods of higher inflow rates in the simulations.

Additionally, turning traffic is simulated and modelled to not decelerate in order to perform a turn. If this would be the case, the deceleration could cause spillback conflicts to occur, increasing the mean density for the same inflow rate. As proposed by [Doole et al. \(2021\)](#), this effect can be coped with by adding altitude layers specifically for turning, in a three-dimensional airspace design.

Despite the limitations discussed above, the analytical model demonstrated the ability to estimate the relationship between the mean inflow rate and the mean density, similar to the free-flow section of a Macroscopic Fundamental Diagram, for a two-dimensional orthogonal grid network. Therefore, the derived model provides an effective tool to predict the effect of several design parameters on the capacity of constrained urban airspace. In addition, the high level of accuracy validates utilisation of the theoretical maximum density defined by [Assumption 3](#) to compare different constrained urban airspace designs, similar to the maximum density defined for general airspace by the CAMDA model ([Sunil et al., 2018a](#)).

8.2. Effect of airspace design parameters

The analytical model’s accuracy level allows for quantifying the effect of several relevant airspace design parameters. Based on the parameters used in the equations, three design parameters are observed to affect the capacity metrics used in this paper: the cruise speed V_{NR} , the horizontal separation requirement S_h , and the number of flow alternations at an intersection. The look-ahead time, when sufficiently long, does not affect the relationship between the mean inflow rate and the mean density. Additionally, the effect of the airspace structure, including, for example, the distance between intersections and the airspace area, is discussed.

When increasing the cruise speed, aircraft can travel a longer distance in the same time window, lowering the resulting mean density. Moreover, the mean delay per aircraft decreases as more aircraft can cross an intersection in the same time window. The analytical model predicts that the mean inflow rate scales proportionally with the cruise speed V_{NR} when keeping the mean density constant. Furthermore, the maximum stable density defined by [Assumption 3](#) remains unchanged. Hence, an increase in cruise speed is predicted to result in a proportional increase in the maximum inflow rate and capacity.

For the horizontal separation requirement S_h , this proportionality is not present. Instead, the analytical model predicts that enlarging S_h reduces capacity. When S_h is increased, the maximum number of aircraft that fit within an airway is lowered, while the crossing times of aircraft are extended. Both factors increase the mean delay per aircraft, reducing the maximum inflow rate and the maximum density.

Furthermore, capacity is predicted to be increased when the flows coming from the two upstream legs of an intersection alternate less frequently, as was stated in [Assumption 1](#). Because the time window between two subsequent aircraft is increased by a factor $\sqrt{2}$ when the upstream flow is alternated (on an orthogonal intersection, see Eqs. [\(11\)](#) and [\(16\)](#)), capacity is decreased for each alternation. Therefore, a system limiting the number of alternations, for example, a traffic signal, may increase capacity. However,

表4
三个实验的分析与模拟容量对比，以最大稳定密度和稳定流入率组合形式呈现。

Experiment	Analytical		Simulated	
	Max. $\overline{N}_{i,W,R}$	Max. $\overline{q}_i$ s ⁻¹	Max. $\overline{N}_{i,W,R}$	Max. $\overline{q}_i$ s ⁻¹
Base	353	0.96	259	0.78
0.5 V_{NR}	353	0.48	249	0.39
2 S_h	181	0.45	93	0.28

实验中确实如此。不过，这在假设3中已有预测，因为随机效应意味着网络在低于用于分析预测的平均流量的流量下就可能变得不稳定。

8. 比较与综合

在本研究中，开发了一个关于平均流入率与平均密度关系的纯分析性自由流模型，以研究和量化影响受限城市空域容量的因素。进行了模拟以验证分析模型的准确性。本节讨论分析模型的准确程度，随后讨论几个空域设计参数对其容量的影响。

8.1. 分析性自由流模型验证

分析模型与模拟结果的比较表明模型准确性很高，准确率高于97%。分析模型能紧密预测平均流入率与平均密度之间关系的形状以及数值。不准确之处在于在所有三个实验中平均密度的高估不到 2.5%。

在第7节中确定了两个局限性。第一个局限性是平均密度的轻微高估。如前所述，这是不平衡交叉路口平均延误高估的结果。第二个局限性是分析模型预测的最大理论密度高估了模拟运行中遇到的最大稳定密度。这可归因于模拟中较高流入率的随机时段。

为模拟转弯交通，假设其不减速。实际减速可能导致回溢冲突，在相同流入率下增加平均密度。如杜尔等人（2021年）所提议，三维空域设计中可通过为转弯添加专门高度层应对此影响。

尽管存在上述局限性，但分析模型展示了估计二维正交网格网络中平均流入率与平均密度之间关系的能力，类似于宏观基本图的自由流部分关系。因此，所推导的模型为预测几个设计参数对受限城市空域容量的影响提供了一个有效工具。此外，高精度验证了利用假设3定义的理论最大密度来比较不同受限城市空域设计的合理性，类似于CAMDA模型（苏尼尔等人，2018a）为一般空域定义的最大密度。

8.2. 空域设计参数的影响

分析模型的准确程度允许量化几个相关空域设计参数的影响。基于方程中使用的参数，**观察到三个设计参数会影响本文中使用的容量指标：巡航速度 V_{NR} 、水平间隔要求 S_h 以及交叉路口的流量交替次数。**前瞻时间足够长时，不会影响平均流入率与平均密度之间的关系。此外，还讨论了空域结构的影响，例如交叉路口之间的距离和空域面积。

当提高巡航速度时，飞机在相同时间窗口内可以飞行更长距离，从而降低由此产生的平均密度。此外，每架飞机的平均延误减少，因为更多飞机可以在相同时间窗口内通过交叉路口。分析模型预测，在保持平均密度不变的情况下，平均流入率与巡航速度 V_{NR} 成比例缩放。此外，假设3定义的最大稳定密度保持不变。因此，预计巡航速度的提高将导致最大流入率和容量成比例增加。

对于水平间隔要求 S_h ，不存在这种比例关系。相反，分析模型预测增大 S_h 会降低容量。当 S_h 增加时，适合在一条航线上的飞机最大数量会减少，而飞机的穿越时间会延长。这两个因素都会增加每架飞机的平均延误，从而降低最大流入率和最大密度。

此外，如假设1所述，当交叉路口两个上游路段的流量交替频率降低时，预计容量会增加。因为当上游流量交替时（在正交交叉路口，见方程(11)和(16)），后续两架飞机之间的时间窗口会增大 $\sqrt{2}$ 倍，每次交替容量都会降低。因此，例如交通信号灯等限制交替次数的系统可能会增加容量。然而，

this system extends the mean queue length, increasing the chance of overflow. In this study, no exact model for an intersection with an alternation limiter is derived. Hence, the break-even point for the benefits of this system cannot be predicted. It is recommended to quantify this break-even point in further research.

Lastly, the structure of the airspace itself has a significant impact. When the distance between two neighbouring intersections is decreased, the probability of overflow increases, resulting in a reduced capacity. Contrarily, if this change results from an enlarged number of intersections and airways in the area, the capacity increases as the proportion of flyable airspace increases. When keeping the airspace area equal, doubling the number of airways in each direction leads to an increase by a factor of 2.1 in capacity.

Concluding, for airspace design, capacity can be optimised by lowering the separation requirement and by increasing the cruise speed, taking into account the necessary safety levels. Additionally, increasing the density of airways within an airspace increases the maximum capacity as well. However, the minimum separation requirement should be considered when increasing the airway density, as the separation requirement circle should not overlap with adjacent airways, leading to head-on conflicts between aircraft in adjacent airways.

9. Conclusions and disucussion

The traffic density of small aerial vehicles within the highly constrained, low-altitude urban airspace is expected to increase significantly in the near future. An analytical approach is taken to increase understanding of the factors that influence the capacity of constrained urban airspace. The emergent behaviour of aircraft in constrained urban airspace, such as queuing and local hotspots, yields the existing analytical models for general airspace invalid. Therefore, in this paper, a new mathematical model is derived to estimate the relationship between the mean inflow rate and the mean density, similar to the free-flow section of a Macroscopic Fundamental Diagram, for a two-dimensional orthogonal grid network. By performing three simulation experiments involving more than 700,000 flights, the level of accuracy of the mathematical model is shown to be higher than 98.5%. This high level confirms the ability of the model to closely predict the shape and magnitude of the relationship between the mean inflow rate and the mean density in the two-dimensional orthogonal grid network.

In addition, the high level of accuracy of the analytical model allows quantitative prediction of the effects of several relevant airspace design parameters on capacity. Notable findings showed that an increase in cruise speed leads to a proportional increase in the maximum network inflow rate while the maximum density is unaffected. Contrarily, enlarging the minimum separation distance reduces capacity. Additionally, turning traffic on balanced intersections does not impact the capacity more than through traffic. If routing with additional turns means that a shorter route is possible, efficiency may be increased. Lastly, a metric was defined describing a theoretical maximum density, which can rank different constrained urban airspace designs on capacity.

Furthermore, in a stable environment, the model predicts that the constrained urban airspace can destabilise when the maximum capacity of a single intersection only is reached. Additionally, for intersections operating at near-maximum capacity, measures limiting flow alternation, such as traffic signals, potentially improve the efficiency of an intersection of airways.

All in all, the results demonstrate how the derived analytical model provides an effective tool to predict the effect of several design parameters on the capacity of constrained urban airspace. Therefore, the analytical method offered by this work can be one of the tools in the toolbox of the future urban airspace designer. Moreover, this model lays the foundation for the derivation of more expanded models, including the altitude dimension and non-orthogonal or non-two-way intersections, enabling an increased understanding of the constrained urban airspace.

There are some further open questions which can be addressed in future work. First, in this paper, a two-dimensional model is derived. However, in a realistic airspace design, the vertical dimension cannot be neglected. The model approach presented in this paper can form a basis for this three-dimensional analysis. By substituting the vertical equivalents of the horizontal variables, e.g., the vertical separation requirement instead of S_h and the vertical speed equivalents of V_{NR} and V_{WR} , the model is expected to be valid for a single vertical plane, two-dimensional airspace. It is recommended to investigate whether the three-dimensional constrained urban airspace can subsequently be modelled as the sum of intersections in either the vertical or the horizontal plane, or whether a separate model extension needs to be derived for the third dimension. Consider that model accuracy is expected to deteriorate when interactions between the vertical and horizontal manoeuvres exist.

Furthermore, the derived model is validated for orthogonal intersections with up to two upstream legs only. In general, more complicated intersections exist with, for example, more than two upstream legs or non-orthogonal legs. It is hypothesised that the model can accurately predict the relationship between the mean inflow rate and the mean density at more complicated intersections, considering a few adjustments. Specifically, the red time r and the cycle time $\bar{c}_u$ need to be adjusted to include all crossing traffic and their respective crossing times. Further research is required to test this hypothesis.

Additionally, several limitations need to be considered for the findings in this paper. This study neglected meteorological effects, which may harm overall capacity. Nevertheless, the meteorological effects can be approximated using, e.g., an increased horizontal separation requirement S_h . Another limitation is the static demand (in terms of network inflow rate) assumed in both the analytical model and the simulations. In reality, airspace demand changes from time to time in urban networks (Geroliminis and Daganzo, 2008). Depending on the size of the airspace and rate of change of aircraft inflow rate, this may imply that during a ‘rush hour’, busy parts of the airspace can congest for a limited time without destabilising the entire airspace (in contrast to Assumption 3). When demand decreases at the end of the rush hour, the congestion gradually disappears. Note that the energy storage of drones is limited, making it plausible that congestion is to be prevented at all times. Also, various other forms of stochasticity in demand (e.g., origin and destination variations over time) and heterogeneity in UAV type should ultimately be included.

该系统延长了平均队列长度，增加了溢出的可能性。在本研究中，未推导出带有交替限制器的交叉路口的精确模型。因此，无法预测该系统效益的盈亏平衡点。建议在进一步研究中对该盈亏平衡点进行量化。

最后，空域本身的结构有重大影响。当两个相邻交叉路口之间的距离减小时，溢出的概率增加，导致容量降低。相反，如果这种变化是由于该区域内交叉路口和航线数量增加所致，随着可飞行空域比例的增加，容量会增加。当保持空域面积不变时，每个方向的航线数量翻倍会使容量增加2.1倍。

总之，对于空域设计，在考虑必要安全水平的情况下，可通过降低间隔要求和提高巡航速度来优化容量。此外，增加空域内航线的密度也会增加最大容量。但是，增加航线密度时应考虑最小间隔要求，因为间隔要求圆不应与相邻航线重叠，否则会导致相邻航线中的飞机发生对头冲突。

9. 结论与讨论

预计在不久的将来，高度受限的低空城市空域内小型飞行器的交通密度将显著增加。本文采用分析方法来增进对影响受限城市空域容量的因素的理解。受限城市空域中飞机的涌现行为，如排队和局部热点，使得现有的一般空域分析模型无效。因此，本文推导了一个新的数学模型，用于估计二维正交网格网络中平均流入率与平均密度之间的关系，类似于宏观基本图的自由流部分。通过进行涉及超过700,000次飞行的三次模拟实验，该数学模型的准确率高于98.5%。这一高准确率证实了该模型能够紧密预测二维正交网格网络中平均流入率与平均密度之间关系的形状和大小。

此外，分析模型较高的准确率使得能够对几个相关空域设计参数对容量的影响进行定量预测。显著的表现表明，巡航速度的增加会导致最大网络流入率成比例增加，而最大密度不受影响。相反，增大最小间隔距离会降低容量。此外，平衡交叉路口的转弯交通对容量的影响并不比直行交通更大。如果带有额外转弯的路由意味着可以采用更短的路线，则效率可能会提高。最后定义了一个描述理论最大密度的指标，该指标可以对不同受限城市空域设计的容量进行排序。

在稳定环境下，该模型预测当仅达到单个交叉路口最大容量时，受限城市空域可能不稳定。对于接近最大容量运行的交叉路口，限制流量交替的措施如交通信号可提高航线交叉路口效率。

总体而言，结果表明所推导的分析模型如何提供了一个有效的工具来预测几个设计参数对受限城市空域容量的影响。因此，本文提供的分析方法可以成为未来城市空域设计师工具包中的工具之一。此外，该模型为推导更扩展的模型奠定了基础，包括高度维度以及非正交或非双向交叉路口，从而能够增进对受限城市空域的理解。

未来工作中还有一些有待解决的开放性问题。首先，本文推导了一个二维模型。然而，在实际空域设计中，垂直维度不能被忽视。本文提出的模型方法可以为这种三维分析奠定基础。通过用垂直变量替代水平变量的等效量，例如用垂直间隔要求替代 S_h ，以及用 V_{NR} 和 V_{WR} 的垂直速度等效量，预计该模型对于单个垂直平面、二维空域是有效的。建议研究三维受限城市空域随后是否可以建模为垂直或水平平面中交叉路口的总和，或者是否需要为第三维度推导单独的模型扩展。考虑到当垂直和水平机动之间存在相互作用时，预计模型准确率会下降。

此外，所推导的模型仅在最多有两条上游路段的正交交叉路口进行了验证。一般来说，存在更复杂的交叉路口，例如有两条以上的上游路段或非正交路段。据推测，考虑一些调整后，该模型可以准确预测更复杂交叉路口的平均流入率与平均密度之间的关系。具体而言，需要调整红色时间 r 和周期时间 $\bar{c}_u$ ，以纳入所有交叉交通及其各自的交叉时间。需要进一步研究来检验这一假设。

此外，本文的研究结果还需要考虑几个局限性。本研究忽略了气象影响，这可能会损害整体容量。然而，气象影响可以通过例如增加水平间隔要求 S_h 来近似。另一个局限性是分析模型和模拟中都假设了静态需求（就网络流入率而言）。在现实中，城市网络中的空域需求会随时间变化（Geroliminis和Daganzo, 2008）。根据空域的大小和飞机流入率的变化率，这可能意味着在“高峰时段”，空域的繁忙部分可能会在有限的时间内拥堵而不会使整个空域不稳定（与假设3相反）。当高峰时段结束需求下降时，拥堵会逐渐消失。请注意，无人机的能量存储有限，因此始终防止拥堵是合理的。此外，最终应纳入需求中的各种其他形式的随机性（例如，随时间的起点和终点变化）以及无人机类型的异质性。

CRediT authorship contribution statement

Michiel J.M. Aarts: Conceptualization, Investigation, Formal analysis, Writing – original draft. **Joost Ellerbroek:** Conceptualization, Supervision, Reviewed and edited the draft. **Victor L. Knoop:** Conceptualization, Supervision, Reviewed and edited the draft.

Data availability

Code can be found in the cited github repository.

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CRediT 作者贡献声明

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数据可用性

代码可在所引用的github存储库中找到。

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